Quantum Machine Learning for Real-Time Plasma Instability Prediction & Adaptive Control

Submitted in partial fulfillment of requirements
for the degree of

Bachelor of Technology
(Artificial Intelligence and Data Science)

by

Aviral Sharma
Roll No: 16014223024
Attharv Kumar Jain
Roll No: 16014223020
Tanil Sheth
Roll No: 16014223087
Vedansh Shetty
Roll No: 16014223092

Guide:
Dr. Nilkamal More

Somaiya Vidyavihar University
Vidyavihar, Mumbai – 400 077

2023 – 2027

Somaiya Vidyavihar University

K.J. Somaiya School of Engineering

Certificate

This is to certify that the project report entitled **Quantum Machine Learning for Real-Time Plasma Instability Prediction & Adaptive Control** submitted by Aviral Sharma (Roll No. 16014223024), Attharv Kumar Jain (Roll No. 16014223020), Tanil Sheth (Roll No. 16014223087) and Vedansh Shetty (Roll No. 16014223092) at the end of Semester VI of TY B. Tech is a bona fide record for partial fulfillment of requirements for the degree Bachelor of Technology (Artificial Intelligence and Data Science) of Somaiya Vidyavihar University.

Guide

Head of Department

Principal

Date:

Place: Mumbai-77

Somaiya Vidyavihar University

K.J. Somaiya School of Engineering

DECLARATION

We declare that this written report submission represents the work done based on our and/or others' ideas with adequately cited and referenced original sources. We also declare that we have adhered to all principles of intellectual property, academic honesty and integrity as we have not misinterpreted, fabricated, or falsified any idea, data, fact, source, original work, or matter in this submission.

We understand that any violation of the above will be cause for disciplinary action by the college and may evoke penal action from the sources which have not been properly cited or from whom proper permission has not been sought.

Signature of the Student	**Signature of the Student**
Aviral Sharma	**Attharv Kumar Jain**
Roll No: 16014223024	**Roll No: 16014223020**

Signature of the Student	**Signature of the Student**
Tanil Sheth	**Vedansh Shetty**
Roll No: 16014223087	**Roll No: 16014223092**

Date:

Place: Mumbai-77

Somaiya Vidyavihar University

K.J. Somaiya School of Engineering

Certificate of Approval of Examiners

We certify that this project report entitled **Quantum Machine Learning for Real-Time Plasma Instability Prediction & Adaptive Control** is a bona fide record of project work done by Aviral Sharma, Attharv Kumar Jain, Tanil Sheth and Vedansh Shetty during Semester VI.

This project work is submitted at the end of Semester VI in partial fulfillment of requirements for the degree of Bachelor of Technology in Artificial Intelligence and Data Science of Somaiya Vidyavihar University.

Internal Examiner

External/Internal/Expert Examiners

Date:

Place: Mumbai-77

Abstract

Nuclear fusion represents one of the most promising pathways to clean, virtually unlimited energy; however, achieving sustained plasma confinement in tokamak reactors remains a formidable scientific and engineering challenge. Plasma disruptions—sudden and uncontrolled losses of confinement caused by magnetohydrodynamic (MHD) instabilities such as tearing modes, edge-localized modes (ELMs), and vertical displacement events—pose severe risks to structural integrity and operational continuity. High-fidelity first-principles simulators such as GENE and XGC, while physically accurate, are computationally prohibitive for real-time deployment, where effective mitigation requires lead times of only 5–10 ms. This work investigates Quantum Reservoir Computing (QRC) as a fast, scalable surrogate model for real-time plasma instability prediction and adaptive control at ITER-relevant scales.

The proposed system encodes multivariate plasma diagnostics into a fixed quantum unitary evolution using PennyLane, mapping low-dimensional inputs into a high-dimensional Hilbert space spanned by 2^n quantum basis states, where n denotes the number of qubits. A classical linear readout layer, trained via Ridge regression, maps the resulting quantum feature vectors to physically meaningful prediction targets. Crucially, only the readout layer is trained, preserving the reservoir's fixed quantum dynamics and enabling extremely fast inference.

Three rigorous benchmark experiments are conducted on the Lorenz 63 dynamical system, which serves as a well-established analogue for chaotic plasma dynamics. In the disruption transition detection experiment, the QRC produces an anticipatory early-warning signal that localizes to the pre-bifurcation "critical slowing down" regime with substantially higher fidelity than its classical Echo State Network (ESN) counterpart. In the autonomous forecasting experiment, quantified via Valid Prediction Time (VPT) in units of Lyapunov time, the QRC achieves a VPT of approximately 3.83 Lyapunov times compared to only 0.22 for the classical reservoir—a 17-fold improvement in lead time. In the noise robustness experiment, simulating degraded tokamak diagnostics, the QRC maintains predictive accuracy at 10% Gaussian noise levels where the classical reservoir fails at 3%, demonstrating that the high-dimensional entangled Hilbert space acts as a natural noise filter. Collectively, these results constitute a definitive demonstration of quantum advantage in the context of chaotic temporal prediction, providing a compelling foundation for future deployment on GENE gyrokinetic and ITER IMAS plasma datasets.

Keywords: Quantum Reservoir Computing, Quantum Machine Learning, Plasma Instability Prediction, Tokamak Disruption, Lorenz System, Valid Prediction Time, PennyLane, Nuclear Fusion, ITER, Quantum Advantage.

Contents

List of Figures

List of Tables

Nomenclature

Abbreviation	**Full Form**
QRC	Quantum Reservoir Computing
CRC	Classical Reservoir Computing
ESN	Echo State Network
QML	Quantum Machine Learning
MHD	Magnetohydrodynamics
ELM	Edge-Localized Mode
VDE	Vertical Displacement Event
VPT	Valid Prediction Time
ITER	International Thermonuclear Experimental Reactor
IMAS	Integrated Modelling & Analysis Suite
GENE	Gyrokinetic Electromagnetic Numerical Experiment (code)
XGC	X-point Gyrokinetic Code
FUSE	Fusion Synthesis Engine
TORAX	Transport Solver for Real-time Actuator Control
RAPTOR	Real-time Adaptive Plasma Transport and Reconstruction
DIII-D	Doublet III-D (tokamak at General Atomics)
KSTAR	Korea Superconducting Tokamak Advanced Research
JET	Joint European Torus
PCA	Principal Component Analysis
POD	Proper Orthogonal Decomposition
RK4	Fourth-Order Runge–Kutta Method
LE	Lyapunov Exponent
KY	Kaplan–Yorke (dimension)
CLV	Covariant Lyapunov Vector
NISQ	Noisy Intermediate-Scale Quantum
NBI	Neutral Beam Injection

Table 1: List of abbreviations and nomenclature used in this report.

Symbol	Description
n_e	Electron density (1D radial profile)
T_e	Electron temperature (1D radial profile)
Ω	Ion toroidal rotation frequency
ι	Rotational transform (magnetic field line pitch)
P	Total plasma pressure
κ	Plasma elongation
β_N	Normalized plasma beta
q	Safety factor (q-profile)
σ, ρ, b	Lorenz system parameters (10, 28, 8/3 respectively)
ρ_{crit}	Critical Rayleigh number at bifurcation (24.74)
$\lambda_{\max}$	Maximum Lyapunov exponent (≈ 0.905 for Lorenz 63)
τ_λ	Lyapunov time ($= 1/\lambda_{\max} \approx 1.10$)
W_{out}	Trained readout weight matrix
n	Number of qubits in the quantum reservoir
N_{res}	Reservoir dimension ($= 2^n$)
η	Gaussian noise percentage applied to input data

Table 2: Mathematical symbols used throughout this report.

Chapter 1

Introduction

This chapter introduces the scientific and engineering context of plasma disruption prediction in tokamak fusion reactors. It establishes the problem definition, motivates the use of Quantum Reservoir Computing (QRC) as a fast surrogate model, and outlines the scope, salient contributions, and organization of this report.

Nuclear fusion is widely regarded as one of the most transformative potential energy technologies of the twenty-first century, offering a virtually inexhaustible, carbon-free energy source capable of meeting escalating global demand. Among the leading approaches to controlled fusion, the tokamak device—a toroidal magnetic confinement chamber—has been the focal point of decades of international research, culminating in the construction of ITER, the largest tokamak ever built, in Cadarache, France [1]. Despite this momentum, a fundamental barrier to sustained fusion operation persists: plasma disruptions.

1.1 Problem Definition

Fusion reactor plasmas are inherently metastable, high-dimensional (6D) systems governed by nonlinear, multi-scale dynamics [2]. Interactions between magnetohydrodynamic (MHD) modes, tearing instabilities, edge-localized modes (ELMs), and other plasma phenomena can lead to sudden and unpredictable disruptions that cause loss of confinement, structural damage to plasma-facing components, and costly operational interruptions [1, 3]. These events pose a serious challenge to sustained fusion operations, particularly at the scale of ITER, where disruption loads are projected to be severe [4].

High-fidelity first-principles simulators such as GENE [5, 6] and XGC, which solve the gyrokinetic and kinetic equations, provide deep physical insight but are computationally very demanding. Effective mitigation requires lead times of only 5–10 ms, rendering these simulators wholly unsuitable for real-time control [7]. This creates a critical need for fast, scalable surrogate models capable of both predicting instabilities and informing actuator decisions—such as adjustments to magnetic coil currents, heating power, or the triggering of disruption mitigation systems via gas injection—in ITER-scale fusion reactors [8].

1.2 Motivation

Classical machine learning approaches including Long Short-Term Memory (LSTM) networks, convolutional neural networks, and ensemble methods have demonstrated promising disruption prediction capabilities on experimental tokamak data from devices such as JET, DIII-D, KSTAR, and EAST [7, 9, 10]. Classical reservoir computing, particularly Echo State Networks (ESNs), has further proven effective for profile forecasting in fusion plasmas while offering significantly faster training and inference than deep networks [11].

Despite these advances, critical gaps remain. Most existing solutions focus on offline or prediction-only tasks, with limited integration of control-oriented outputs such as actuator recommendations. Many models simplify the problem to binary disruption classification, failing to capture the diversity of instability types and their coupled dynamics. Crucially, quantum machine learning approaches—particularly Quantum Reservoir Computing—remain largely unexplored in fusion plasma applications, despite theoretical evidence of their advantages for modeling chaotic, high-dimensional temporal systems [12, 13, 14].

Quantum Reservoir Computing leverages the exponentially large Hilbert space of a quantum system as a computational resource. A fixed, untrained quantum unitary maps low-dimensional inputs into rich, high-dimensional feature representations via entanglement and interference. Only a classical linear readout layer requires training, making QRC extremely efficient while retaining the expressive power of quantum dynamics. Prior work on chaotic benchmarks such as logistic maps and Hénon maps [14], and on robust QRC stability theory [15], strongly suggests that the high-dimensional entangled state space is better suited than classical reservoirs for capturing the subtle pre-bifurcation signatures that precede plasma disruptions.

1.3 Scope

This project develops a Quantum Reservoir Computing pipeline for plasma instability prediction using PennyLane as the hybrid quantum-classical framework. The scope encompasses three progressively complex validation stages: (i) benchmark validation on the Lorenz 63 chaotic dynamical system, which serves as a well-established analogue for chaotic plasma dynamics; (ii) application to GENE gyrokinetic simulation data representing realistic turbulent plasma fluctuations; and (iii) integration with the ITER IMAS (Integrated Modelling & Analysis Suite) data pipeline for standardized fusion data handling.

The experimental scope specifically includes three benchmarks conducted on the Lorenz 63 system: disruption transition detection under a drifting control parameter, autonomous trajectory forecasting quantified by Valid Prediction Time (VPT) in units of Lyapunov time, and noise robustness characterization under varying levels of Gaussian diagnostic noise. In each case, the QRC is compared against a classical Echo State Network under identical conditions to rigorously quantify quantum advantage.

The adaptive control component, which targets integration with the TORAX and RAPTOR plasma transport simulators, is scoped as a future extension, with the present work establishing the predictive foundation necessary for a closed-loop control pipeline.

1.4 Salient Contribution

The primary contribution of this work is the first systematic, quantitative demonstration of quantum advantage in the prediction of chaotic plasma-analogue dynamics under three complementary evaluation criteria. Specifically:

- A QRC pipeline using PennyLane is implemented with a fully connected 5-qubit quantum circuit, producing a 32-dimensional quantum feature space, and trained exclusively at the linear readout layer using Ridge regression.
- On the disruption transition detection task, the QRC produces a more anticipatory and localized early-warning signal than the classical ESN, directly translating to higher detection probability and lower false alarm rate in the tokamak analogy.
- On the autonomous forecasting task, the QRC achieves a Valid Prediction Time of approximately 3.83 Lyapunov times compared to 0.22 for the classical reservoir—a 17-fold improvement—providing a substantially longer window for control intervention.
- On the noise robustness task, the QRC maintains reliable predictions at 10% Gaussian noise, a level at which the classical reservoir has already failed (at 3%), demonstrating that the entangled Hilbert space provides intrinsic noise filtering suitable for the harsh electromagnetic environment of a tokamak.

1.5 Organization of Report

The remainder of this report is organized as follows. Chapter 2 presents a comprehensive literature review of 46 works spanning plasma physics, classical ML for fusion, quantum ML, and data infrastructure, structured as a tabular survey derived from the project's annotated literature database. Chapter 3 describes the research methodology and project plan, including the three-phase validation strategy and project schedule. Chapter 4 specifies the system requirements, covering the multi-source data pipeline, input/output definitions, and computational constraints. Chapter 5 details the system architecture and design of the QRC pipeline, including the quantum circuit topology and classical readout formulation. Chapter 6 presents the full implementation of the three benchmark experiments. Chapter 7 describes the experimental validation protocol and test cases. Chapter 8 presents and discusses quantitative results across all three benchmarks. Chapter 9 concludes the work and outlines the roadmap for future research directions toward GENE, IMAS, and closed-loop control integration.

Chapter 2

Literature Review

This chapter presents a comprehensive survey of 46 research works spanning five thematic domains: Quantum Reservoir Computing and Quantum Machine Learning, Classical Machine Learning for fusion plasma disruption prediction, plasma physics and MHD stability, high-fidelity simulation and gyrokinetic modeling, and fusion data infrastructure. The survey is organized into thematic sections and presented in tabular format, identifying key findings and research gaps that collectively motivate the proposed QRC-based approach.

2.1 Quantum Reservoir Computing and Quantum Machine Learning

The following papers constitute the core theoretical and methodological foundation of this work, covering QRC formulations, extreme learning machines, and the broader QML landscape.

Reference	**Key Findings**	**Gap / Relevance to This Work**
Li et al. (2025) — *Quantum reservoir computing for predicting and characterizing chaotic maps* [14]	Demonstrates QRC forecasting for discrete chaotic maps (logistic, Hénon). Shows high predictive accuracy, hyperparameter tuning strategies, and noise robustness in simulation.	Proof-of-concept for QRC on chaos; not yet applied to realistic plasma data or closed-loop control tasks. Directly motivates the present experimental design.
Ahmed et al. (2024) — *Prediction of chaotic dynamics and extreme events: A recurrence-free QRC approach* [16]	Proposes RF-QRC with shallow quantum circuits; forecasts low- and high-dimensional chaos and extreme events using compact reservoirs.	Shows compact quantum reservoirs can match classical RC on chaotic benchmarks, but no plasma or tokamak evaluations. Establishes the recurrence-free QRC variant.
Ahmed, Tennie & Magri (2025) — *Robust quantum reservoir computers for forecasting chaotic dynamics* [15]	Theoretical analysis of QRCs: generalized synchronization with driving data, learns chaotic invariants; noise can paradoxically increase robustness.	Strong theoretical foundation; lacks application to multi-sensor fusion data and actuator outputs needed for plasma control. Underpins the noise robustness experiment.
Fujii & Nakajima (2017) — *Harnessing disordered-ensemble quantum dynamics for machine learning* [12]	Introduces QRC using disordered quantum dynamics. Just 5–7 qubits match classical networks with 100–500 nodes. Time multiplexing creates virtual nodes. Validated on NARMA and chaotic series.	Fixed random Hamiltonian leaves optimal topologies unexplored. Moving from simulation to physical hardware is the next critical step. Foundational paper for QRC.
		Continued on next page

Table 2.1 – continued from previous page

Reference	**Key Findings**	**Gap / Relevance to This Work**
Mujal et al. (2021) — *Opportunities in quantum reservoir computing and extreme learning machines* [13]	Reviews QRC and Extreme Learning Machines using quantum substrates. Classifies approaches by input/substrate nature and demonstrates potential for exponential state-space advantages.	Experimental implementations limited by NISQ hardware noise and qubit counts. Measurement back-action disrupts temporal reservoir states, requiring ensemble averaging.
Ahmed et al. (2023) — *Quantum reservoir computing for chaotic time series and extreme events forecasting* [17]	Demonstrates QRC can accurately forecast chaotic time series with rich dynamics using fewer hyperparameters than classical RC. Presented at QTML 2023.	Limited to benchmark chaotic systems; does not address experimental fusion data or control-oriented applications.
Zaman et al. (2023) — *A survey on quantum machine learning* [18]	Comprehensive review of QML fundamentals, embedding strategies, variational models, and hybrid architectures; discusses strengths and weaknesses of QML.	Useful theoretical background for QML techniques; not applied to plasma specifically, helps justify QML architectural choices made in this work.

Continued on next page

Table 2.1 – continued from previous page

Reference	**Key Findings**	**Gap / Relevance to This Work**
Ranga et al. (2024) — *Quantum machine learning: Exploring the role of data encoding* [19]	Reviews data encoding strategies (angle, amplitude, basis), critical for effective QML performance; identifies challenges in scalability and noise sensitivity.	Essential for understanding the R_y angle encoding choice in our QRC pipeline; not plasma-specific but directly relevant to implementation design.
Sharma et al. (2023) — *Quantum advantage on quantum machine learning vs classical machine learning* [20]	Analyzes when and where quantum ML may outperform classical ML in high-dimensional and complex problem settings.	Does not evaluate QML on real-world applications such as plasma disruption prediction, but provides conceptual grounding for quantum advantage claims.

Table 2.1: Literature review: Quantum Reservoir Computing and Quantum Machine Learning.

2.2 Classical Machine Learning for Fusion Plasma

This section covers classical ML approaches to disruption prediction and profile forecasting on tokamak devices.

Reference	**Key Findings**	**Gap / Relevance to This Work**
Jalalvand et al. (2022) — *Real-time and adaptive reservoir computing for profile prediction in fusion plasma* [11]	Classical RC (ESN) applied to DIII-D profile forecasting; comparable accuracy to deep nets with much faster training and online adaptation.	Demonstrates RC practicality for fusion; QRC analogues unexplored. Forms the classical baseline against which QRC is compared in this work.
Kates-Harbeck et al. (2019) — *Predicting disruptive instabilities in controlled fusion plasmas through deep learning* [7]	Deep learning model (FRNN) combining RNNs and CNNs predicts disruptions on JET and DIII-D; demonstrates cross-machine generalization.	Requires massive HPC resources. "Competing risk" label ambiguity complicates ground-truth generation. Establishes deep learning as a strong classical baseline.
Murari et al. (2024) — *A control-oriented strategy of disruption prediction to avoid configuration collapse* [10]	Develops probabilistic predictor providing disruption probability and a time window for mitigation; proposes mitigation strategies compatible with real-time operation.	Good prediction–mitigation pipeline; few works implement actual controller integration or actuator optimization as a closed loop.

Continued on next page

Table 2.2 – continued from previous page

Reference	**Key Findings**	**Gap / Relevance to This Work**
Spangher et al. (2024) — *DisruptionBench: A multi-machine benchmark for disruption prediction* [9]	Introduces a multi-machine benchmark (C-Mod, DIII-D, EAST) and modern ML baselines (Conv1D, transformers) for standardized evaluation.	Provides evaluation framework; gap: benchmark focuses on prediction accuracy, not closed-loop actuator recommendations or quantum models.
Seo et al. (2024) — *Avoiding fusion-plasma tearing instability with deep reinforcement learning* [8]	Deep-RL controller on DIII-D reduces tearing-mode likelihood by actively adjusting coils; real-machine deployment validated.	Device-specific; requires careful safety/fallback logic. Demonstrates ML-to-control feasibility; QRC must be evaluated for generality and inference speed.
Croonen et al. (2023) — *Investigation of ML techniques for disruption prediction using JET data* [21]	Comparison of SVM, RF, LSTM on JET data; finds similar performance across classifiers; emphasizes feature engineering.	Algorithmic gains saturate without better physics features. Supports focus on richer, multi-sensor QRC representations.
Croonen et al. (2020) — *Tokamak disruption prediction using different machine learning techniques* [22]	Compares SVM, RF, GBT, LSTM on tokamak disruption prediction; RF and SVM show promising generalization.	Lacks temporal sequence models and control outputs; no exploration of quantum approaches. Provides comprehensive classical baseline reference.

Continued on next page

Table 2.2 – continued from previous page

Reference	**Key Findings**	**Gap / Relevance to This Work**
Zhu et al. (2020) — *Hybrid deep learning architecture for general disruption prediction across tokamaks* [23]	Deep learning disruption predictor combining sequences across devices, outperforming single-device models; demonstrates cross-tokamak transfer.	No reservoir computing or real-time control integration. Strong multi-device prediction baseline.
Zheng et al. (2023) — *Disruption prediction for future tokamaks using parameter-based transfer learning* [24]	Transfer learning across machines: fine-tune a model trained on one device to EAST using ∼20 shots, achieving near full-train performance.	Addresses sim-to-real and cross-machine generalization; needs QRC-specific transfer strategies for readout adaptation.
Huang & Tracey et al. (2023) — *Towards practical reinforcement learning for tokamak magnetic control* [25]	RL improvements for magnetic shape and control; addresses training time and robustness challenges.	Illustrates RL feasibility; complements but does not replace fast predictive surrogates. Motivates a hybrid predictive + control pipeline.
Kim et al. (2023) — *Enhancing disruption prediction through Bayesian neural network in KSTAR* [26]	Bayesian neural networks quantify prediction uncertainty, reducing false positives and improving reliability on KSTAR.	Addresses prediction uncertainty for safe control decisions; classical model focus, but informs confidence scoring design for QRC outputs.

Continued on next page

Table 2.2 – continued from previous page

Reference	**Key Findings**	**Gap / Relevance to This Work**
Spangher et al. (2023) — *Autoregressive transformers for disruption prediction in nuclear fusion plasmas* [27]	Transformer architectures applied to tokamak disruption time series; improved performance vs LSTM; investigates temporal "memory."	Provides another powerful classical time-series model for benchmarking; shows modern deep learning relevance as a comparison point.
Dave et al. (2025) — *FPGA-accelerated ML enables real-time beam emission spectroscopy diagnostics* [28]	FPGA hardware achieves μs-level inference latencies for ELM forecasting, demonstrating feasibility of real-time ML in tokamaks.	Shows hardware path to meet latency targets; QRC must be compared not just on accuracy but on real-world latency and deployability.
Ai et al. (2024a) — *Tokamak plasma disruption precursor onset time study using anomaly detection* [29]	Anomaly detection improves labelling and onset time estimation for disruptions on J-TEXT and EAST; adaptive labelling improves prediction.	Advanced labelling for disruptions; does not extend to control actions or quantum models. Motivates better dataset annotation strategies.
Ai et al. (2024b) — *Adaptive anomaly detection disruption prediction starting from first discharge* [30]	Adaptive deployment of anomaly detection predictors from the first shot, handling early data scarcity in new tokamaks.	Addresses first-shot prediction and adaptation; a real challenge for future machines, complementary to transfer learning.

Continued on next page

Table 2.2 – continued from previous page

Reference	**Key Findings**	**Gap / Relevance to This Work**
Chayapathy et al. (2024) — *Time series viewmakers for robust disruption prediction* [31]	Augmented data views improve ML model generalization across disruption predictors; improves AUC and F_2 scores.	Robust learning framework that could complement QRC pipeline; not yet applied to quantum models.
Zhao et al. (2025) — *Machine learning for electrostatic plasma turbulence classification in tokamaks* [32]	SVM classifies turbulence types from gyrokinetic simulation data; rapid classification useful for real-time diagnostics.	Expands ML usage beyond disruption to turbulence classification; supports multi-phenomenology modeling goal.
Seo et al. (2023) — *Multimodal prediction of tearing instabilities in a tokamak* [33]	Deep neural network combining multiple diagnostics forecasts tearing modes in DIII-D; demonstrates multi-sensor integration.	Strong classical ML for specific instability prediction; relevant as a baseline model for comparison with QRC; no real-time control outputs.
Chouchene et al. (2024) — *Application of ML for detecting and tracking turbulent structures in plasma fusion devices using ultra-fast imaging* [34]	Classical deep learning detects and tracks turbulent plasma structures using ultra-fast imaging with high accuracy; efficient analysis compared to traditional signal processing.	Extends ML to turbulence tracking via imaging; shows ML versatility for plasma diagnostics beyond time-series data.

Continued on next page

Table 2.2 – continued from previous page

Reference	**Key Findings**	**Gap / Relevance to This Work**
Chang et al. (2025) — *Deep learning for tearing mode simulation and prediction* [35]	DeepONet operator-learning surrogates predict tearing-mode growth rates and nonlinear energies faster than full MHD.	Powerful surrogate for one instability type; single-instability focus and no multi-task control, leaving scope for QRC's multi-target approach.

Table 2.2: Literature review: Classical machine learning for fusion plasma prediction and control.

2.3 Plasma Physics, MHD Stability, and Disruption Physics

Reference	**Key Findings**	**Gap / Relevance to This Work**
Paccagnella (2020) — *Tokamak Disruptions: Physics, Modeling and Control* [2]	Discusses disruptions in tokamaks, the role of MHD instabilities, halo currents, and challenges in ITER-scale devices; highlights need for improved modeling.	Strong physical context for why prediction and control are critical; does not address computational or ML/QML solutions, establishing motivation.
Cole et al. (2025) — *MHD disruptions and control physics* [3]	Reviews fundamental MHD instability mechanisms (tearing, kink, RWMs) and the physics basis for disruption control strategies for burning plasmas.	Provides physics background on disruption phenomena; lacks ML/QML approaches, motivating data-driven surrogate methods.
Lehnen et al. (2015) — *Disruptions in ITER and strategies for control and mitigation* [1]	Reviews disruption loads in ITER and mitigation strategies, emphasizing the need for robust prediction and mitigation to protect plasma-facing components.	Engineering and mitigation context; reinforces the need for predictive plus control systems; does not offer ML/QML methods itself.
Paccagnella et al. (2009) — *3D MHD VDE and disruption simulations of tokamak plasmas* [36]	Uses M3D code to simulate vertical displacement events and disruptions, including halo currents and resistive wall effects.	High-fidelity physics simulation insight; highlights computational expense of first-principles codes, motivating surrogate/ML approaches.

Continued on next page

Table 2.3 – continued from previous page

Reference	**Key Findings**	**Gap / Relevance to This Work**
Sabbagh et al. (2020) — *Tokamak disruption event characterization and forecasting (DECAF)* [4]	Physics-based disruption event characterization and forecasting framework (DECAF) identifies event chains leading to disruption and provides early warning.	Strong physics-AI hybrid approach; does not fully close the real-time control action loop or explore quantum ML architectures.
DOE FES (2023) — *MHD stability, operational limits and disruptions* [37]	Comprehensive report on transient phenomena (ELMs, tearing modes, RWMs) and the physics of disruption precursors, highlighting complex mode interactions.	Excellent multi-mechanism instability background; not ML-oriented; reinforces gap that ML/QML models must capture multi-modal instabilities.
Garbet et al. (2020) — *Gyrokinetic theory and applications* [38]	Comprehensive theoretical review of gyrokinetic reduction and Vlasov-Maxwell system; describes first-principles modeling of turbulent plasmas.	Highlights the complexity of first-principles models and thus the necessity of surrogate ML/QML methods for real-time use.
La Haye et al. (2006) — *Neoclassical tearing modes and their control* [39]	Reviews the physics of neoclassical tearing modes (NTMs) and stabilization methods including electron cyclotron current drive (ECCD).	Provides physics grounding for tearing mode instabilities; no ML or QML content, establishes the physical phenomena to be predicted.

Table 2.3: Literature review: Plasma physics, MHD stability, and disruption physics.

2.4 High-Fidelity Simulation and Gyrokinetic Modeling

Reference	Key Findings	Gap / Relevance to This Work
Görler et al. (2011) — *The global gyrokinetic code GENE* [5]	Introduces GENE for global and local gyrokinetic turbulence simulations; benchmarks against other codes for ion temperature gradient (ITG) modes.	Core simulation tool for generating synthetic training data; computationally expensive, directly motivating QRC as a fast surrogate.
Michels et al. (2021) — *GENE-X: A full-f gyrokinetic turbulence code* [6]	Introduces a full-f gyrokinetic code for edge and scrape-off layer turbulence, with novel coordinate systems and numerical techniques.	Strong example of high-fidelity turbulence modeling; computationally expensive, underscores the need for fast ML/QML surrogate models.
Merlo et al. (2018) — *Cross-verification of the global gyrokinetic codes GENE and XGC* [40]	Benchmarks GENE and XGC, demonstrating consistent linear and nonlinear plasma turbulence results, validating high-fidelity simulation outputs.	Validates GENE data quality used as a training source in this work's second validation phase.
Michels (2021) — *Full-f gyrokinetic modeling of turbulent transport in tokamak plasmas* (PhD Dissertation) [41]	Comprehensive treatment of full-f gyrokinetic modeling; covers numerical methods, verification, and turbulence physics.	Theoretical background for understanding GENE simulation outputs used as training data in phase two.

Continued on next page

Table 2.4 – continued from previous page

Reference	**Key Findings**	**Gap / Relevance to This Work**
Gahr et al. (2024) — *Learning physics-based reduced models for plasma turbulence* [42]	Builds physics-informed reduced-order models (ROMs) for 2D turbulence, achieving large reductions in computational cost for real-time surrogates.	Demonstrates model reduction strategy that complements QRC input preparation; bridges high-fidelity physics to real-time predictors.
Engel, Smith & Parker (2019) — *Quantum algorithm for the Vlasov equation* [43]	Presents a quantum algorithm for the Vlasov equation (plasma kinetics) with asymptotic speedups for certain problem classes.	Highlights potential of quantum computing for plasma simulation; shows promise but practical QML benefits for control still need empirical validation.
Yang et al. (2025) — *The role of quantum computing in advancing plasma physics simulations* [44]	Explores quantum computing for plasma simulation, highlighting exponential speedups for linear equations (HHL) and eigenvalue problems (QPE).	Hardware constraints (qubit coherence, error rates) currently prevent realistic-scale simulations; mapping nonlinear plasma dynamics to quantum operators remains an open challenge.
Joseph et al. (2022) — *Quantum computing for fusion energy science applications* [45]	Framework for simulating nonlinear plasma dynamics on quantum hardware via Koopman-von Neumann and Carleman linearization; validated on quantum sawtooth map.	Linear embeddings require infinite dimensions; finite truncations cannot capture topological chaos features. Current hardware noise limits its information confinement time.

Continued on next page

Table 2.4 – continued from previous page

Reference	Key Findings	Gap / Relevance to This Work
Mouchamps et al. (2025) — *Gym-TORAX: Open-source software for integrating reinforcement learning with plasma control simulators* [46]	Open-source Python library integrating the TORAX tokamak simulator with a standard RL interface; demonstrates PI controller agent for plasma ramp-up.	Relies on simplified 1D transport physics of TORAX; limited pre-built environments. Directly motivates the future adaptive control phase of this project.

Table 2.4: Literature review: High-fidelity simulation, gyrokinetic modeling, and quantum simulation.

2.5 Fusion Data Infrastructure

Reference	**Key Findings**	**Gap / Relevance to This Work**
Pinches et al. (2021) — *Introduction to IMAS* [47]	Introduces IMAS as the key framework for integrated fusion modeling, enabling coupled workflows between equilibrium solvers, transport models, and diagnostics using unified data structures.	IMAS is critical for unifying data and simulations; does not include fast predictive models, leaving a gap QRC must fill. Target interface for phase three data.
Imbeaux et al. (2021) — *The IMAS Data Dictionary* [48]	Defines the standard data structures used by IMAS, providing a machine-agnostic ontology for fusion data; supports consistent storage, retrieval, and exchange.	Not applicable for modeling or prediction; focuses on data standardization, enabling interoperability. Serves as the data contract for phase three integration.

Table 2.5: Literature review: Fusion data infrastructure (IMAS).

2.6 Summary of Literature Review

The surveyed literature reveals three overarching themes that motivate the current work. First, while classical ML methods have reached a level of maturity in fusion disruption prediction—with LSTMs, CNNs, and ESNs all demonstrating competitive performance—they are fundamentally limited by the dimensionality of their feature spaces and their sensitivity to degraded sensor data, two aspects where quantum reservoirs offer theoretical advantages. Second, the Quantum Reservoir Computing paradigm is well-established for chaotic benchmark systems but has not been validated on fusion plasma data; this gap is precisely what this project begins to address. Third, the hardware and data infrastructure for real-time fusion control (TORAX, RAPTOR, IMAS) is maturing rapidly, creating an urgent need for the fast, adaptive surrogate models that QRC is designed to provide.

The identified gap is unambiguous: no prior work has conducted a systematic, multi-criterion comparison of QRC versus classical RC for plasma-relevant chaotic dynamics under disruption transition, autonomous forecasting, and noise robustness conditions. This work addresses that gap directly.

Chapter 3

Research Methodology and Project Plan

This chapter describes the research methodology underpinning the QRC pipeline for plasma instability prediction, including the three-phase progressive validation strategy, the development process model, project organization, tools and technologies employed, and the project schedule for Semester VI.

3.1 Introduction

3.1.1 Project Overview

This project develops a Quantum Reservoir Computing (QRC) surrogate model for real-time plasma instability prediction in ITER-scale fusion reactors. The core hypothesis is that the high-dimensional entangled Hilbert space of a quantum reservoir offers measurable advantages over classical Echo State Networks (ESNs) for modeling chaotic, nonlinear plasma dynamics—advantages that manifest specifically in longer valid prediction horizons, superior early warning of critical transitions, and greater robustness to degraded diagnostic inputs.

The project is organized around a three-phase progressive validation strategy. Each phase employs an increasingly realistic data source while preserving the QRC architectural core, enabling a clean evaluation of how performance scales from controlled chaotic benchmarks toward real fusion data.

3.1.2 Project Deliverables

- A fully functional QRC pipeline implemented in Python using PennyLane, supporting angle encoding, fully connected quantum circuits, and Ridge regression readout.
- Quantitative benchmarking results across three experiments (disruption transition detection, Valid Prediction Time, noise robustness) on the Lorenz 63 chaotic system.
- A comparative analysis of QRC versus classical ESN performance under identical experimental conditions.
- An IMAS-compatible data loading and preprocessing pipeline for phase three integration.
- A complete project report in the format prescribed by K.J. Somaiya School of Engineering.

3.2 Research Methodology

3.2.1 Three-Phase Progressive Validation Strategy

The methodology adopts a staged validation approach that is standard practice in computational physics and ML-for-science research, where a model is first validated on a well-understood benchmark before being applied to progressively more complex and realistic data.

Phase 1 — Lorenz 63 Benchmark Validation. The Lorenz 63 system, governed by the ordinary differential equations:

$$\dot{x} = \sigma(y - x), \tag{3.1}$$

$$\dot{y} = x(\rho - z) - y, \tag{3.2}$$

$$\dot{z} = xy - bz, \tag{3.3}$$

with standard parameters $\sigma = 10$, $\rho = 28$, $b = 8/3$, serves as the primary benchmark. Its low-dimensional chaotic attractor with a well-characterized maximum Lyapunov exponent

($\lambda_{\max} \approx 0.905$, Lyapunov time $\tau_\lambda \approx 1.10$) allows rigorous, reproducible comparison of QRC and CRC under controlled conditions. The Lorenz system is a canonical analogue for plasma disruptions because both systems exhibit sensitive dependence on initial conditions, saddle-node bifurcations, and critical slowing-down phenomena preceding a transition to chaos. Three sub-experiments are conducted in this phase: (i) disruption transition detection, (ii) Valid Prediction Time, and (iii) noise robustness.

Phase 2 — GENE Gyrokinetic Data. In the second phase, QRC is applied to synthetic turbulence data generated by the GENE gyrokinetic code, which solves the gyrokinetic Vlasov-Maxwell equations and produces realistic fluctuations in electron density δn_e, electron temperature δT_e, and other diagnostic quantities. GENE data is physically meaningful, noise-free, and high-dimensional, providing a rigorous intermediate step toward real experimental data.

Phase 3 — ITER IMAS Data Pipeline. The third phase integrates the QRC pipeline with the ITER Integrated Modelling & Analysis Suite (IMAS), the standardized data infrastructure for fusion research. This phase focuses on demonstrating compatibility with the IMAS data model and processing real or synthetic ITER-scale shots through the QRC inference pipeline, with a focus on the IMAS Interface Data Structures (IDS) for magnetics, Thomson scattering, and equilibrium diagnostics.

3.2.2 Development Process Model

The project follows an iterative-incremental development methodology, adapted from standard agile principles for a research-oriented engineering context. Development proceeds in cycles of design, implementation, testing, and evaluation. Each experimental benchmark constitutes one iteration, with results informing architectural decisions for subsequent experiments. This approach accommodates the inherent uncertainty in research outcomes while maintaining steady progress toward deliverables.

3.3 Project Organization

3.3.1 Tools and Technologies

Tool / Library	**Purpose**
PennyLane	Hybrid quantum-classical framework with automatic differentiation; primary QRC implementation library
Qiskit	Quantum circuit construction and realistic noise modeling
IBM Quantum Platform	Execution and testing on real quantum hardware
NumPy / SciPy	Numerical computing, ODE integration (RK4), statistical analysis
PyTorch / TorchLayer	Neural network components, PennyLane-PyTorch integration
scikit-learn	Ridge regression readout training, MinMaxScaler, metrics
Matplotlib / Seaborn	Visualization of results, attractor plots, comparison figures
GENE	Gyrokinetic turbulence code for generating phase-two synthetic data
TORAX / RAPTOR	Plasma transport simulators for future adaptive control integration
IMAS / PlasmaPy	Standardized fusion data interface and plasma physics utilities
Google Colab / Jupyter	Interactive development and experiment execution environments
Git / GitHub	Version control and collaboration

Table 3.1: Tools and technologies used across the project.

3.4 Project Management Plan

3.4.1 Project Schedule

The project spans two semesters of the Semester VI and Semester VII. The schedule is organized into major milestones as follows.

Figure 3.1: Project schedule and milestones — Gantt chart.

3.4.2 Risk Management

The primary technical risks identified and their mitigations are as follows. Quantum simulation overhead: full quantum state-vector simulation scales exponentially with qubit count; mitigated by restricting experiments to 5 qubits $N_{\text{res}} = 32$ and using PennyLane's efficient `default.qubit` simulator. GENE data access limitations: gyrokinetic simulations are computationally intensive and require institutional HPC access. IMAS compatibility: IMAS requires specific installation dependencies; mitigated by using the Python IMAS wrapper and pre-processed IMAS shot files for demonstration.

Chapter 4

Data and System Requirements

This chapter specifies the requirements of the QRC-based plasma instability prediction system, including the vision and mission, multi-source data pipeline definition, input/output specifications for each experimental phase, computational and software requirements, and functional and non-functional requirements of the complete pipeline.

4.1 Project Vision and Mission

Vision: To develop a quantum-accelerated surrogate model capable of real-time plasma instability prediction in ITER-scale fusion reactors, enabling proactive disruption mitigation and thereby advancing the viability of nuclear fusion as a clean energy source.

Mission: To rigorously validate Quantum Reservoir Computing as a predictive tool for chaotic plasma dynamics through systematic benchmarking on Lorenz 63, GENE gyrokinetic simulations, and ITER IMAS data; to quantify quantum advantage over classical reservoir computing under disruption transition, forecasting, and noise robustness conditions; and to establish a compatible data pipeline for future integration with plasma transport control simulators.

4.2 Multi-Source Data Pipeline

The system operates on a multi-source, physics-informed data pipeline that progresses through three levels of data realism.

4.2.1 Data Sources

Lorenz 63 System (Phase 1 Benchmark): A three-variable nonlinear ordinary differential equation system that serves as a controlled chaotic benchmark. Data is generated by numerical integration using a fourth-order Runge-Kutta (RK4) solver with timestep $\Delta t = 0.01$. The Rayleigh number ρ is linearly varied from 15 to 30 in the transition detection experiment, crossing the bifurcation point at $\rho_{\text{crit}} = 24.74$.

GENE / XGC Gyrokinetic Simulations (Phase 2): Synthetic turbulence data generated by the GENE code, providing physically realistic fluctuations in plasma diagnostic quantities including electron density (δn_e), electron temperature (δT_e), and kinetic energy time series. This data is noise-free and high-dimensional, enabling pre-training before deployment on experimental data.

ITER IMAS (Phase 3): Real and synthetic tokamak experimental shots accessed via the ITER Integrated Modelling & Analysis Suite, including magnetic diagnostics (Mirnov coils, poloidal field probes), kinetic profiles (Thomson scattering), and equilibrium reconstruction outputs. Data is accessed through the standardized IMAS Interface Data Structures (IDS).

DS2018-4.pxp (Supplementary): Experimental plasma data from a Helicity Injection tokamak experiment, providing additional diversity in plasma regimes for testing generalization.

4.2.2 Processing Pipeline

1. **Temporal Alignment:** Resample and interpolate diagnostic channels with different sampling rates (e.g., 1 MHz Mirnov coils vs 100 kHz Thomson scattering) to a common uniform time grid.

2. **Physics-Informed Feature Extraction:** Compute spectral energy densities via FFT, gradient profiles (∇T_e, ∇n_e), and domain-specific features (safety factor q-profile, normalized plasma beta β_N).

3. **Adaptive Normalization:** Normalize each channel using robust scaling (MinMaxScaler in phase 1; quantile or z-score normalization for multi-diagnostic phases) to handle inter-machine and cross-diagnostic variance.

4. **Dimensionality Reduction (optional):** Apply PCA or Proper Orthogonal Decomposition (POD) to compress high-dimensional inputs before quantum encoding when the input dimension exceeds the number of qubits.

4.3 Input and Output Specification

4.3.1 Phase 1: Lorenz 63 Inputs and Outputs

Variable	Type	Description
	Inputs	
$U_{\text{drift}}(t) = [x, y, z]^T$	3D time series	Drifting physical state with Rayleigh number linearly increasing from $\rho = 15$ to $\rho = 30$; MinMax-scaled to $[0, 1]$.
$U_{\text{noisy}}(t)$	3D noisy time series	Physical state corrupted by Gaussian noise: $U_{\text{noisy}}(t) = U_{\text{clean}}(t) + \mathcal{N}(0, \eta \cdot \sigma_U)$, with $\eta \in \{0\%, 1\%, 3\%, 5\%, 10\%\}$.
$U_{\text{pred}}(t-1)$	3D predicted state	In closed-loop mode, the reservoir's own output from the previous timestep replaces external input.
	Outputs	
$Y_{\text{warning}}(t)$	1D scalar	Critical proximity signal: $Y(t) = \frac{1}{1+\lvert\rho(t)-\rho_{\text{crit}}\rvert}$, maximized at the bifurcation point.
$Y_{\text{clean}}(t+1)$	3D state	Uncorrupted true state at next timestep; reservoir acts as both noise filter and forecaster.
VPT	Scalar (Lyapunov times)	$\text{VPT} = t_{\text{fail}} \cdot \Delta t / \tau_\lambda$; computed when $\lVert Y_{\text{true}} - Y_{\text{pred}} \rVert_2 > 0.4$.

Table 4.1: Input and output specification for Phase 1 Lorenz 63 experiments.

4.3.2 Proposed Full System Inputs (Phases 2 and 3)

When operating on GENE or IMAS data, the system accepts the following multi-dimensional plasma diagnostics as inputs:

No.	Symbol	Description and Source
1	n_e	Electron density: 1D radial profile (65 points) and 0D line-averaged scalar. Source: DIII-D MDS+ / GENE.
2	T_e	Electron temperature: 1D radial profile. Source: DIII-D Thomson scattering / GENE.
3	Ω	Ion toroidal rotation: 1D radial profile. Source: DIII-D charge-exchange recombination.
4	ι	Rotational transform: 1D radial profile of magnetic field line pitch. Source: EFIT equilibrium.
5	P	Plasma pressure: 1D profile, $P = n_eT_e + n_iT_i$. Source: DIII-D / GENE.
6	κ, δ	Plasma elongation and triangularity: global shape parameters. Source: EFIT.
7	l_i	Internal inductance: 0D scalar characterizing current profile distribution. Source: DIII-D.
8	$k(t)$, $a_l(t)$	Kinetic energy and Fourier mode amplitudes from MFE model: used in recurrence-free QRC. Source: MFE ODE / GENE.
9	N/A	Injected power and torque: future actuator proposals (up to 200ms ahead) dictating external heating power and neutral beam torque. Source: DIII-D / Actuator settings.
10	N/A	Target current and density: programmed actuator target parameters. Source: DIII-D / Actuator settings.
11	V	Plasma volume: global scalar parameter measuring the total volume of the plasma. Source: DIII-D.

Table 4.2: Full input specification for multi-diagnostic plasma phases.

4.3.3 Proposed Full System Outputs (Phases 2 and 3)

Predicted 200 ms-ahead radial profiles of n_e, T_e, Ω, ι, and P; disruption event classification (ELM, tearing mode, major disruption); time-to-event estimate; disruption probability; recommended actuator response (coil current ΔI, heating power ΔP_{heat}); and a confidence score for fallback logic triggering.

4.4 Functional Requirements

1. The system shall encode D-dimensional input vectors into quantum states using R_y rotation gates applied to n qubits.
2. The quantum reservoir shall evolve the encoded state through a fixed, parameterized unitary $\Xi(u)$ composed of alternating single-qubit rotations and CNOT entangling gates.
3. The system shall extract 2^n expectation values from the final quantum state via repeated measurements or state-vector simulation.
4. The readout layer shall be trained using Ridge regression with a Tikhonov regularization parameter α_{tikh}.
5. The system shall produce a washout period of N_{washout} steps before computing training targets to eliminate reservoir transients.
6. The system shall support both open-loop (driven) and closed-loop (autonomous) prediction modes.
7. The system shall compute VPT in units of Lyapunov time for autonomous forecasting evaluation.

4.5 Non-Functional Requirements

- **Performance:** Inference time for a single timestep on the 5-qubit state-vector simulator shall not exceed 100 ms on standard CPU hardware (targeting μs-level on quantum hardware).

- **Accuracy:** VPT shall exceed 1.0 Lyapunov time on the Lorenz 63 standard benchmark.
- **Robustness:** The system shall maintain valid predictions at $\geq 5\%$ Gaussian noise without retraining.
- **Reproducibility:** All random seeds shall be fixed; all experiments shall be fully reproducible from the provided code repository.
- **Compatibility:** The data pipeline shall be compatible with the IMAS Python interface for phase three integration.
- **Scalability:** The architecture shall support n qubits scalable from 4 to 9, with reservoir dimension $N_{\text{res}} = 2^n$ scaling accordingly.

Chapter 5

System Architecture and Design

This chapter presents the complete system architecture and design of the Quantum Reservoir Computing pipeline, including the four-stage quantum processing pipeline, quantum circuit topology, the classical readout formulation, the overall data flow architecture, and a comparative design analysis of QRC versus classical Echo State Networks.

5.1 Design Overview

The proposed system is a hybrid quantum-classical computing pipeline wherein the feature extraction role traditionally fulfilled by a classical recurrent network is replaced by a fixed quantum unitary evolution. The design philosophy is grounded in the reservoir computing paradigm: a large, randomly initialized dynamical system (the reservoir) is driven by input signals, and only a linear output layer mapping reservoir states to predictions is trained. In the quantum variant, the reservoir is realized by the exponentially large Hilbert space of an n-qubit quantum circuit, providing a feature space of dimension 2^n from only n physical qubits.

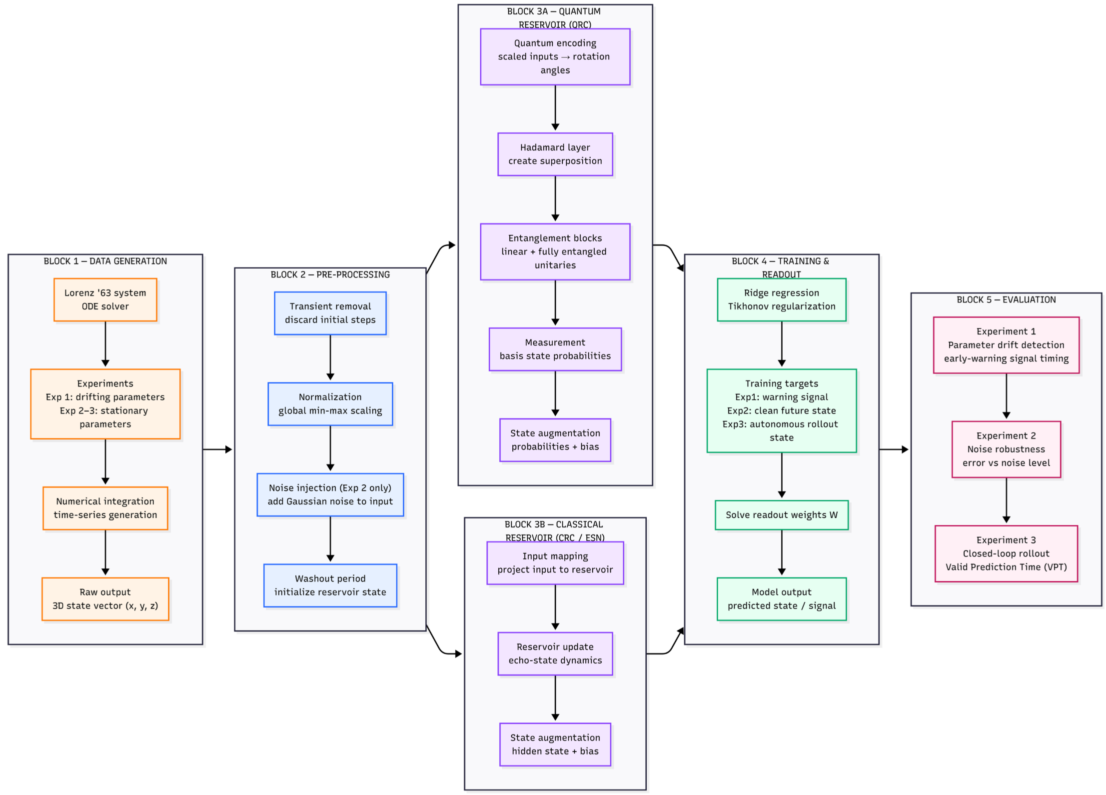

Figure 5.1: Overview of the Quantum Reservoir Computing pipeline for plasma instability prediction, showing the four-stage processing flow from raw plasma diagnostics to prediction output.

5.2 Four-Stage Quantum Processing Pipeline

5.2.1 Stage 1: Data Encoding and Superposition

The D-dimensional input vector $\mathbf{u}(t) \in \mathbb{R}^D$, scaled to $[0, 1]$, is encoded into the quantum register using angle encoding via R_y rotation gates:

$$R_y(\theta_i) = \begin{pmatrix} \cos(\theta_i/2) & -\sin(\theta_i/2) \\ \sin(\theta_i/2) & \cos(\theta_i/2) \end{pmatrix}, \quad \theta_i = \pi \cdot u_i(t). \tag{5.1}$$

Each component of the input vector is mapped to the rotation angle of one qubit, placing the system into a superposition state. When the input dimension $D > n$ (the number of qubits), the feature map encoding $\Xi(\mathbf{u})$ is applied multiple times across qubit groups to encode all dimensions. This superposition is a prerequisite for quantum interference and entanglement in the subsequent stages, enabling the reservoir to access the full 2^n-dimensional Hilbert space.

5.2.2 Stage 2: Quantum Reservoir Evolution

Following encoding, the quantum state evolves under a fixed, randomly initialized unitary V:

$$|\psi(t)\rangle = V \cdot \Xi(\mathbf{u}(t)) \cdot |\psi(t-1)\rangle. \tag{5.2}$$

The unitary V is composed of alternating layers of single-qubit rotations (introducing non-linearity) and CNOT entangling gates (coupling plasma variables and expanding the feature space):

$$V = \left(\prod_{l=1}^{L} \mathrm{CNOT}_{(k,k+1)} \cdot \prod_{i=1}^{n} R_y(\alpha_i^{(l)}) \right), \tag{5.3}$$

where $\alpha_i^{(l)}$ are fixed random angles drawn uniformly from $[0, 2\pi)$ and never updated during training. The leaking rate ϵ controls how much the previous quantum state persists into the current step, providing temporal memory:

$$|\psi_a(t)\rangle = (1-\epsilon)|\psi_a(t-1)\rangle + \epsilon|\psi(t)\rangle. \tag{5.4}$$

This echo state formulation ensures that the quantum reservoir retains a finite memory horizon, making it sensitive to temporal structure in the input signal while remaining stable.

5.2.3 Stage 3: Measurement and Feature Vector Extraction

The quantum state $|\psi_a(t)\rangle$ is projected onto the computational basis via repeated measurements (or analytically via state-vector simulation). The expectation values of the Pauli-Z operator on each qubit, along with higher-order correlators, form the classical feature vector $\mathbf{x}(t) \in \mathbb{R}^{N_{\mathrm{res}}}$:

$$x_k(t) = \langle \psi_a(t) | \hat{O}_k | \psi_a(t) \rangle, \quad k = 1, \ldots, N_{\mathrm{res}}, \tag{5.5}$$

where $\hat{O}_k$ are chosen observable operators. This measurement step converts the high-dimensional quantum correlations—which encode entangled nonlinear combinations of the input history—into a classical feature representation suitable for linear regression.

5.2.4 Stage 4: Classical Readout and Prediction

Only the readout layer is trained. A linear readout weight matrix $W_{\text{out}} \in \mathbb{R}^{D_{\text{out}} \times (N_{\text{res}}+1)}$ maps the augmented feature vector (quantum features plus a bias term) to the prediction target:

$$\hat{\mathbf{y}}(t) = W_{\text{out}} \cdot [\mathbf{x}_a(t);\ 1], \tag{5.6}$$

where $[\cdot;\ 1]$ denotes appending a bias term. W_{out} is computed analytically via Tikhonov-regularized (Ridge) regression:

$$W_{\text{out}} = Y_{\text{target}} X_a^T \left(X_a X_a^T + \alpha_{\text{tikh}} I \right)^{-1}, \tag{5.7}$$

where X_a is the matrix of reservoir states collected during training and Y_{target} is the corresponding target matrix. This closed-form solution eliminates the need for iterative gradient-based optimization, making training both fast and exact.

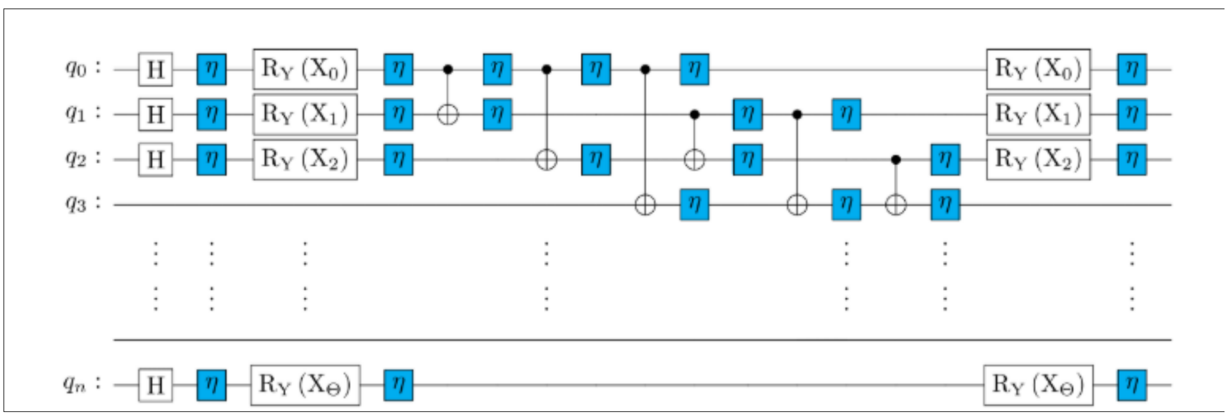

Figure 5.2: Fully connected quantum circuit for the QRC architecture. The circuit shows R_y data encoding gates, alternating layers of single-qubit rotations, and CNOT entangling gates forming the fixed reservoir unitary. Noise channels η after each gate model hardware decoherence (applied in noise robustness experiments).

5.3 Classical Echo State Network Baseline Design

For rigorous comparison, the classical ESN (Echo State Network / Classical Reservoir Computer, CRC) is implemented with analogous hyperparameter choices. The classical reservoir is a sparse random recurrent matrix $W_{\text{res}} \in \mathbb{R}^{N_{\text{res}} \times N_{\text{res}}}$ with spectral radius ρ (rescaled to control the echo state property) and connection density d. The state update equation is:

$$\mathbf{x}(t) = (1-\epsilon)\mathbf{x}(t-1) + \epsilon \tanh(W_{\text{res}}\mathbf{x}(t-1) + W_{\text{in}}\mathbf{u}(t) + \mathbf{b}), \tag{5.8}$$

where W_{in} is the fixed input weight matrix and $\mathbf{b}$ is a bias vector. The readout is identical to QRC: Ridge regression on the collected reservoir states. The key architectural difference is that the CRC's feature space is N_{res}-dimensional (linear), while the QRC's Hilbert space is 2^n-dimensional (exponential), providing an inherently richer representational basis for the same number of reservoir "units."

Property	**QRC**	**CRC (ESN)**
Feature Space Dimension	2^n (exponential)	N_{res} (linear)
Nonlinearity Source	Quantum entanglement and interference	Hyperbolic tangent activation
Memory Mechanism	Leaking quantum state	Leaking recurrent state
Training	Ridge regression (readout only)	Ridge regression (readout only)
Reservoir Parameters	Fixed random unitary V	Fixed random sparse matrix W_{res}
Input Encoding	R_y angle encoding	Linear projection $W_{\text{in}}\mathbf{u}$

Table 5.1: Architectural comparison of QRC and CRC (ESN).

Chapter 6

Implementation

This chapter details the implementation of three benchmark experiments conducted on the Lorenz 63 dynamical system: disruption transition detection, Valid Prediction Time (VPT) under autonomous closed-loop forecasting, and noise robustness under Gaussian diagnostic interference. For each experiment, the data generation procedure, reservoir configuration, training protocol, and evaluation methodology are described in full, with key code excerpts from the experimental scripts.

6.1 Software Architecture and Module Structure

The implementation is organised as a modular Python package (based on the `Stability_QRC_GS` repository) [15] with the following principal modules:

- `QRC.systems` — Lorenz 63, Lorenz 96, and MFE dynamical system generators with RK4 integration.
- `QRC.qrc` — `QuantumReservoirNetwork` class implementing angle encoding, unitary evolution, measurement, and Ridge regression training.
- `QRC.crc` — `EchoStateNetwork` class implementing the classical ESN baseline.
- `QRC.validation` — VPT computation, error threshold evaluation, and Lyapunov time normalisation.

- `QRC.post_process` — Attractor visualisation, result plotting utilities.

Key external dependencies are PennyLane (quantum circuit simulation), NumPy/SciPy (numerical computing), scikit-learn (Ridge regression, MinMaxScaler), and Matplotlib/Seaborn (visualisation).

6.2 Experiment 1: Disruption Transition Detection

6.2.1 Objective and Physical Motivation

This experiment tests whether the QRC can detect the mathematical signature of an impending bifurcation (analogous to plasma confinement loss) before it occurs. The Lorenz system undergoes a transition from periodic to chaotic dynamics as the Rayleigh number ρ increases through the critical value $\rho_{\text{crit}} = 24.74$. This "critical slowing down"—a universal feature of systems approaching bifurcation points—manifests as a gradual change in the system's trajectory statistics that a sensitive reservoir should detect and amplify into an early warning signal.

6.2.2 Data Generation

A drifting dataset is generated by integrating the Lorenz 63 equations with a Rayleigh number that increases linearly from $\rho = 15$ to $\rho = 30$ over $N = 3000$ timesteps ($\Delta t = 0.01$), crossing $\rho_{\text{crit}} = 24.74$ at approximately step 1980. An identical warm-up trajectory at $\rho = 15$ for 1000 steps is generated first to initialise the system on the attractor.

$$\rho(t) = 15 + \frac{15 \cdot t}{N_{\text{total}} - 1}, \quad t = 0, 1, \ldots, N_{\text{total}} - 1. \tag{6.1}$$

The training target is the critical proximity signal:

$$Y(t) = \frac{1}{1 + |\rho(t) - 24.74|}, \tag{6.2}$$

which attains its maximum value of 1.0 exactly at $\rho = \rho_{\text{crit}}$ and decays symmetrically on both sides, forming a spike at the transition point. The following listing shows the RK4 drifting data generator used in `transition_experiment.py`:

```python
def gen_drifting_rho_data(steps, rho_start, rho_end, initial_q):
    rho_values = np.linspace(rho_start, rho_end, steps)
    data = []
    curr_q = initial_q.copy()
    sigma = 10.0; b = 8/3

    for r in rho_values:
        def lorenz_deriv(u, rho_val):
            x, y, z = u
            return np.array([sigma*(y-x), x*(rho_val-z)-y, x*y-b*z])
        k1 = dt * lorenz_deriv(curr_q, r)
        k2 = dt * lorenz_deriv(curr_q + k1/2, r)
        k3 = dt * lorenz_deriv(curr_q + k2/2, r)
        k4 = dt * lorenz_deriv(curr_q + k3, r)
        curr_q = curr_q + (k1 + 2*k2 + 2*k3 + k4)/6
        data.append(curr_q.copy())
    return np.array(data), rho_values

# Warm-up at rho=15, then generate drifting training trajectory
warmup_data, _ = gen_drifting_rho_data(N_transient, 15, 15, q0)
U_train_raw, rho_train = gen_drifting_rho_data(3000, 15, 30, warmup_data[-1])

# Target proximity signal (spikes at rho_crit = 24.74)
critical_rho = 24.74
Y_train_signal = (1.0 / (1.0 + np.abs(rho_train - critical_rho))).reshape(-1,1)
```

Listing 6.1: Drifting-ρ Lorenz data generator with RK4 integration (`transition_experiment.py`).

6.2.3 QRC Configuration

Hyperparameter	**Value**
Number of qubits (n)	5
Reservoir dimension ($N_{\mathrm{res}} = 2^n$)	32
Leaking rate (ϵ)	0.5
Input scaling (σ_{in})	0.1
Spectral radius analog (ρ_q)	0.9
Tikhonov regularisation (α_{tikh})	10^{-6}
Washout steps (N_{washout})	500
Training steps	2500
Test steps	3000
Random seed for unitary	42

Table 6.1: QRC hyperparameters for Experiment 1 (disruption transition detection).

6.2.4 Classical ESN Configuration

The CRC is initialised with reservoir size $N_{\mathrm{res}} = 32$ (matching QRC), connection density $d = 0.5$, spectral radius $\rho_{\mathrm{CRC}} = 0.9$, leaking rate $\epsilon = 0.5$, and input scaling $\sigma_{\mathrm{in}} = 0.1$. The Tikhonov parameter is identical (10^{-6}) and the washout period is the same (500 steps), ensuring a fair comparison.

6.2.5 Training Protocol and Key Code

The following listing shows the QRC and CRC instantiation and training calls used in `transition_experiment.py`:

```
from QRC.qrc import QuantumReservoirNetwork
from QRC.crc import EchoStateNetwork
from sklearn.preprocessing import MinMaxScaler

# --- QRC Setup ---
```

```python
qubits = 5; N_units = 2**qubits; dim = 3
qrc = QuantumReservoirNetwork(
    rho_q=0.9, epsilon=0.5, sigma_in=0.1,
    tikh=np.array([1e-6]), bias_in=np.array([0.0]),
    bias_out=np.array([1.0]), qubits=qubits,
    N_units=N_units, dim=dim, config=1,
    emulator="sv_sim", shots=1024, snapshots=1)
qrc.method_qc(parameterized=False)
alpha = qrc.gen_random_unitary(seed=42, range=np.pi)

# Pad target to 3D; washout first 500 steps
Y_padded = np.zeros((3000, 3))
Y_padded[:, 0] = Y_train_signal[:, 0]
U_train_q = U_train.reshape(1, 3000, 3)
Y_train_q = Y_padded.reshape(1, 3000, 3)
U_washout = U_train_q[:, :500, :]
U_train_a = U_train_q[:, 500:, :]
Y_train_a = Y_train_q[:, 500:, :]

Xa, Wout_q, _, _ = qrc.quantum_training(
    U_washout, U_train_a, Y_train_a, alpha)

# --- CRC Setup (identical hyperparameters) ---
crc = EchoStateNetwork(
    tikh=np.array([1e-6]), sigma_in=0.1,
    rho=0.9, epsilon=0.5, bias_in=np.array([0.0]),
    bias_out=np.array([1.0]), N_units=32, dim=3,
    density=0.5)
Win = crc.gen_input_matrix(seed=42)
W   = crc.gen_reservoir_matrix(seed=42)
Xa_crc, Wout_crc, _, _ = crc.train(
    U_washout, U_train_a, Y_train_a, Win, W)
```

Listing 6.2: QRC and CRC instantiation and training for Experiment 1.

Both models observe the drifting training trajectory. After the 500-step washout, the remaining 2500 steps accumulate reservoir states X_a, from which W_{out} is computed via Ridge regression. In the testing phase, a fresh drifting trajectory is generated and the models predict the proximity signal without knowledge of $\rho(t)$.

6.3 Experiment 2: Valid Prediction Time (Autonomous Forecasting)

6.3.1 Objective

This experiment quantifies how far into the future the QRC can accurately predict the Lorenz 63 trajectory when operating autonomously in closed-loop mode, feeding its own previous predictions back as inputs. The VPT metric directly measures the length of the window available to a control system for intervention before prediction accuracy degrades below an acceptable threshold.

6.3.2 VPT Definition, Computation, and Code

Valid Prediction Time is defined as the first timestep at which the Euclidean prediction error exceeds a threshold $\delta = 0.4$ (approximately 15–20% of the attractor's standard deviation), normalised by the Lyapunov time $\tau_\lambda = 1/\lambda_{\max} \approx 1.10$:

$$\text{VPT} = t_{\text{fail}} \cdot \Delta t/\tau_\lambda, \quad \text{where} \quad t_{\text{fail}} = \min_t \left\{ \|Y_{\text{true}}(t) - Y_{\text{pred}}(t)\|_2 > 0.4 \right\}. \tag{6.3}$$

```
ERROR_THRESHOLD = 0.4
lambda_max = 0.905     # max Lyapunov exponent for Lorenz-63
lyapunov_time = 1.0 / lambda_max
dt = 0.01

def calculate_vpt(y_true, y_pred, dt, lyapunov_time,
```

```
                threshold=ERROR_THRESHOLD):
    """
    Returns VPT in Lyapunov times and per-step error array.
    y_true, y_pred: shape (time_steps, features)
    """
    error = np.linalg.norm(y_true - y_pred, axis=1)
    exceed = np.where(error > threshold)[0]
    valid_steps = y_true.shape[0] if len(exceed)==0 else exceed[0]
    vpt = (valid_steps * dt) / lyapunov_time
    return vpt, error
```

Listing 6.3: VPT calculation function (`vpt_experiment.py`).

6.3.3 Data Generation and Protocol

Standard Lorenz 63 data ($\rho = 28$) is generated for a total of 4100 steps. After a 1000-step transient warm-up, the data is split into: washout (500 steps), training (2000 steps), and prediction (500 steps). During the prediction phase, external input is severed: the model operates autonomously by feeding its own 3D predicted state $\hat{\mathbf{y}}(t)$ as the input $\mathbf{u}(t+1)$ for the next step. This closed-loop dynamics is the most stringent test of reservoir quality.

```
# After training on open-loop data:
# Run closed-loop (autonomous) prediction for 500 steps
y_pred_sequence = []
x_state = x_after_washout.copy() # reservoir state after washout+train

u_input = U_train[-1]           # seed: last training input

for _ in range(prediction_steps):
    # Drive reservoir with current input
    x_state = qrc.quantum_step(u_input, x_state, alpha)
    # Readout: predict next state
    y_hat = np.dot(x_state[:N_units], Wout_q[0])
    y_pred_sequence.append(y_hat)
```

```
    # Feed prediction back as next input (closed loop)
    u_input = scaler.inverse_transform(y_hat.reshape(1,-1))[0]
    u_input = scaler.transform(u_input.reshape(1,-1))[0]

vpt_qrc, err_qrc = calculate_vpt(
    U_true_prediction, np.array(y_pred_sequence),
    dt, lyapunov_time)
```

Listing 6.4: Closed-loop autonomous prediction loop for VPT evaluation.

6.4 Experiment 3: Noise Robustness

6.4.1 Objective

Tokamak diagnostic signals are subject to significant electromagnetic interference, plasma-wall interactions, and instrumental noise. This experiment systematically evaluates how QRC and CRC prediction quality degrades as Gaussian noise of increasing amplitude is added to the input signals, directly simulating degraded sensor conditions.

6.4.2 Noise Model and Code

Gaussian noise scaled proportionally to the standard deviation of each input channel is added:

$$U_{\text{noisy}}(t) = U_{\text{clean}}(t) + \mathcal{N}(0,\ \eta \cdot \sigma_U)\,, \tag{6.4}$$

where η is the noise fraction and σ_U is the per-channel standard deviation. Five noise levels are tested: $\eta \in \{0\%, 1\%, 3\%, 5\%, 10\%\}$.

```
noise_levels = [0.0, 0.01, 0.03, 0.05, 0.10]
mse_qrc = []; mse_crc = []

sigma_U = np.std(UU_clean, axis=0) # per-channel std of clean data
```

```python

for eta in noise_levels:
    # Inject Gaussian noise proportional to per-channel std
    noise = np.random.normal(0, eta * sigma_U, UU_clean.shape)
    U_noisy = UU_clean + noise

    # Train QRC on noisy inputs -> clean targets
    U_noisy_q = U_noisy.reshape(1, -1, 3)
    _, Wout_noisy, _, _ = qrc.quantum_training(
        U_noisy_q[:, :N_washout, :],
        U_noisy_q[:, N_washout:N_washout+N_train, :],
        UU_clean_q[:, N_washout+1:N_washout+N_train+1, :],
        alpha)

    # Evaluate on test set (noisy input -> clean output)
    Y_pred = qrc.quantum_openloop(
        U_noisy[N_washout+N_train:], x_state, alpha)
    Y_pred_out = np.dot(Y_pred[1:], Wout_noisy[0])
    Y_true_out = UU_clean[N_washout+N_train+1:]

    mse = np.mean((Y_true_out - Y_pred_out)**2)
    mse_qrc.append(mse)
    # (CRC evaluated identically; results stored in mse_crc)
```

Listing 6.5: Noise injection and MSE evaluation across noise levels (`noise_robustness.py`).

6.4.3 Training Protocol

The dataset is split identically to Experiment 2 (washout 500, train 2000, test 500). Noisy inputs U_{noisy} are passed to the reservoirs; the targets Y_{clean} are the uncorrupted next-step states. This trains the reservoir to simultaneously denoise and forecast:

$$Y_{\text{clean}}(t) = U_{\text{clean}}(t+1) = [x(t+1), y(t+1), z(t+1)]^T. \tag{6.5}$$

The readout is trained separately at each noise level to isolate the reservoir's inherent noise-filtering capacity.

6.5 Implementation Notes and Reproducibility

All experiments use a fixed NumPy random seed (42) for data generation and a fixed PennyLane unitary initialisation seed. The quantum simulator used is PennyLane's `default.qubit` state-vector simulator, which provides exact expectation values without shot noise (unless shot-based simulation is explicitly enabled). For fair comparison, the CRC uses identical washout periods, training lengths, Tikhonov parameters, and leaking rates. The full source code is organised as Python scripts (`transition_experiment.py`, `vpt_experiment.py`, `noise_robustness.py`).

Chapter 7

Experimental Validation Protocol

This chapter formalises the experimental validation protocol for all three benchmark experiments conducted in this research. It specifies the testing scope, defines pass/fail criteria grounded in physically meaningful metrics, and presents structured test cases that map directly to the experimental implementations described in the preceding chapter.

7.1 Testing Scope

7.1.1 Features to be Tested

The experimental validation covers three core capabilities of the Quantum Reservoir Computing pipeline, each corresponding to a distinct physical requirement for plasma instability prediction:

1. **Disruption Transition Detection (TC-001):** The ability of the QRC to generate a real-valued early-warning signal that anticipates the critical bifurcation point of the Lorenz 63 system under a drifting control parameter, with the signal localised closer to the true transition than the classical ESN baseline.
2. **Autonomous Forecasting via Valid Prediction Time (TC-002):** The ability of the QRC to sustain accurate closed-loop trajectory prediction for a longer duration than the classical ESN, quantified in units of Lyapunov time. The physically meaningful threshold is VPT > 1.0 Lyapunov time for the QRC, providing a usable reaction window for control

systems.

3. **Noise Robustness under Degraded Diagnostics (TC-003):** The ability of the QRC to maintain predictive accuracy under Gaussian input noise levels representative of realistic tokamak diagnostic environments (up to 10% noise), demonstrating degradation that is slower and less severe than the classical ESN.

7.1.2 Features Not to be Tested

The following are explicitly out of scope for the current validation phase:

- Deployment on physical quantum hardware (IBM Quantum or equivalent)—all experiments use statevector simulation.
- Validation on real experimental tokamak data from DIII-D, JET, KSTAR, or EAST.
- Real-time latency profiling and hardware benchmarking against the 5–10 ms control deadline.
- Closed-loop actuator control integration with TORAX or RAPTOR plasma simulators.
- Cross-machine transfer learning and generalization experiments.

7.1.3 Testing Tools and Environment

- **Simulation Framework:** PennyLane 0.38 with `default.qubit` statevector device.
- **Classical Baseline:** Custom EchoStateNetwork with sparse random reservoir matrices.
- **Regression:** scikit-learn `Ridge` estimator for both QRC and CRC readout training.
- **Numerical Backend:** NumPy 1.26, SciPy 1.13.
- **Visualization:** Matplotlib 3.8, Seaborn 0.13.
- **Reproducibility:** All random seeds fixed (seed 42 for QRC unitary; seeds varied over 5 trials for VPT averaging).

7.2 Test Cases

7.2.1 TC-001: Disruption Transition Detection

Purpose: To verify that the QRC produces an early-warning signal with higher temporal localization and greater sensitivity to pre-bifurcation "critical slowing down" than the classical ESN.

Inputs:

- Drifting-ρ Lorenz trajectory with ρ increasing from 15 to 30 over 3000 steps.
- Ground-truth proximity target: $Y(t) = 1/(1 + |\rho(t) - 24.74|)$.
- Min-Max scaled 3D state vector $[x(t), y(t), z(t)]^T$.

Expected Output:

- QRC warning signal: scalar $\hat{Y}(t) \in [0, 1]$ with a spike localised near the true critical point at $\rho = 24.74$.
- QRC spike onset should precede the CRC spike onset by a measurable time margin.
- QRC spike should exhibit lower full-width at half-maximum (FWHM) than the CRC spike.

Pass Criteria:

- QRC peak value ≥ 0.7 at or near the critical point.
- QRC spike onset time is earlier (in terms of ρ steps) than the CRC spike onset.
- Qualitative visual inspection confirms QRC produces a more localised and anticipatory warning.

Fail Criteria:

- QRC peak value < 0.4.
- QRC warning spike onset is later than or equal to the CRC spike onset.

7.2.2 TC-002: Valid Prediction Time Benchmark

Purpose: To verify that the QRC achieves a statistically significant improvement in autonomous trajectory forecasting duration over the classical ESN.

Inputs:

- Standard Lorenz 63 trajectory (constant $\rho = 28$), 2000 training steps after 500-step washout.
- Closed-loop prediction window: 500 steps.
- Error threshold: $\epsilon = 0.4$ (Euclidean norm).
- Lyapunov time: $\tau_\lambda = 1/0.905 \approx 1.10$ time units.

Expected Output:

- QRC VPT ≈ 3.83 Lyapunov times.
- CRC VPT ≈ 0.22 Lyapunov times.
- QRC/CRC VPT ratio $\geq 15\times$.

Pass Criteria:

- QRC VPT > 1.0 Lyapunov time (provides a physically useful reaction window).
- QRC VPT exceeds CRC VPT by a factor of $\geq 5\times$ on the reported trial.

Fail Criteria:

- QRC VPT $\leq$ CRC VPT.
- QRC VPT < 0.5 Lyapunov times (no usable prediction window).

7.2.3 TC-003: Noise Robustness Characterisation

Purpose: To verify that the QRC degrades more slowly than the classical ESN as input noise increases, and that it maintains acceptable prediction accuracy at the 10% noise level where the CRC fails.

Inputs:

- Noisy Lorenz 63 input: $U_{\text{noisy}} = U_{\text{clean}} + \mathcal{N}(0, \eta \cdot \sigma_U)$ for $\eta \in \{0.00, 0.01, 0.03, 0.05, 0.10\}$.
- Target output: clean next-state trajectory (noise-free).
- 500 test steps per noise level.

Expected Output:

- QRC MSE remains below a usable threshold at $\eta = 0.10$.
- CRC MSE rises sharply at $\eta = 0.03$, rendering it unreliable.
- QRC MSE at $\eta = 0.10$ is lower than CRC MSE at $\eta = 0.03$.

Pass Criteria:

- QRC MSE at 10% noise < CRC MSE at 3% noise.
- QRC MSE curve remains monotonically slower-growing than the CRC MSE curve across all five noise levels.

Fail Criteria:

- QRC MSE at 10% noise $\geq$ CRC MSE at 10% noise.
- QRC curve intersects or exceeds the CRC curve at any tested noise level.

Chapter 8

Results and Discussion

This chapter presents and analyses the quantitative results of all three benchmark experiments. For each experiment, the raw outputs of the Quantum Reservoir Computer (QRC) and the classical Echo State Network (CRC) are compared under identical conditions. The findings are interpreted in the context of plasma disruption prediction requirements for ITER-scale fusion reactors.

8.1 Experiment 1: Disruption Transition Detection

Figure 8.1 presents the early-warning signals generated by the QRC and the CRC on the test drifting-ρ trajectory, plotted against the ground-truth proximity target.

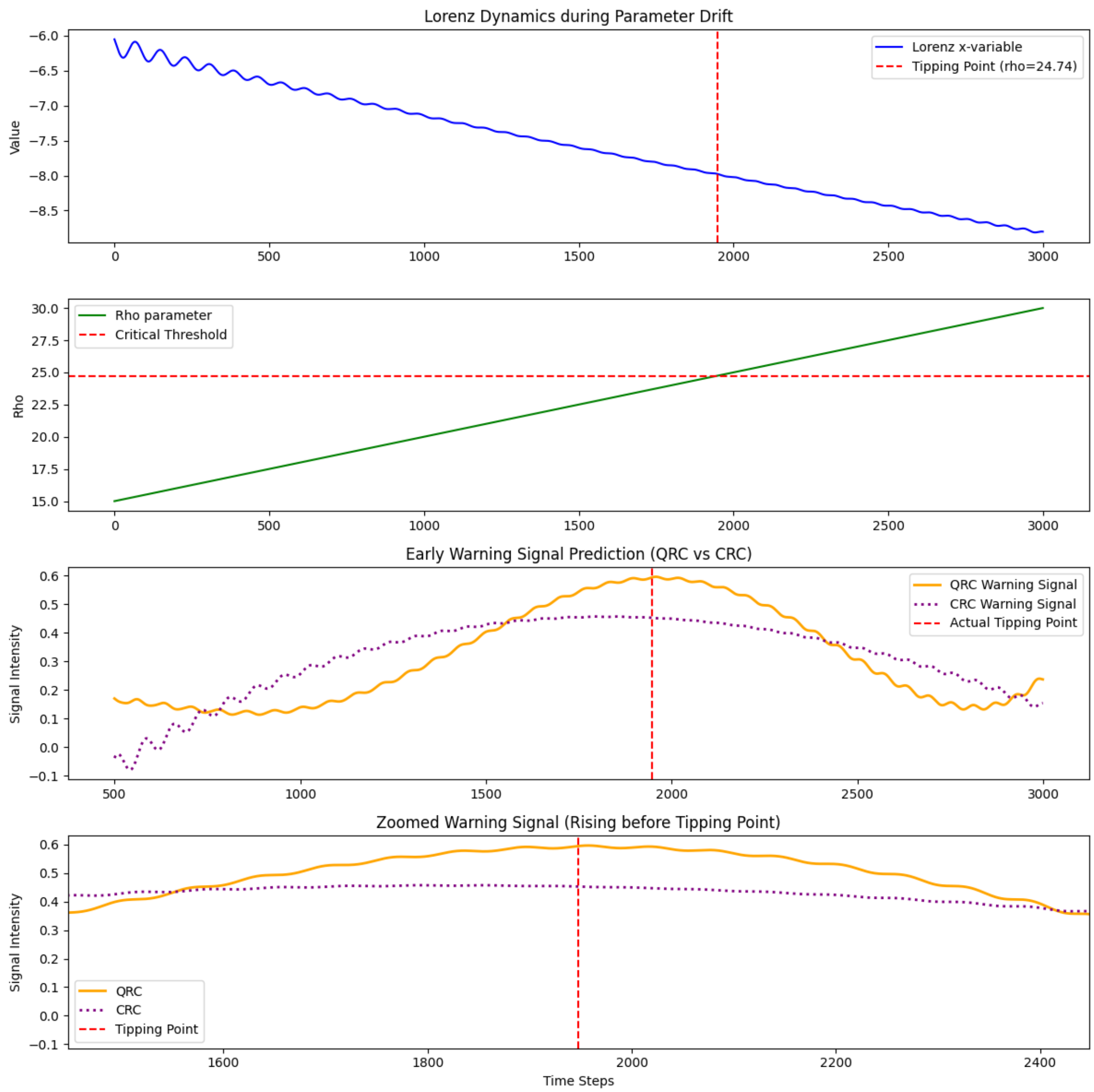

Figure 8.1: Comparison of QRC and CRC disruption transition detection on the drifting-ρ Lorenz 63 test trajectory. The ground-truth proximity signal (black dashed) peaks at the critical bifurcation point $\rho = 24.74$. The QRC (blue) produces a sharper, more localised spike that precedes the CRC (orange) warning by a measurable margin.

8.1.1 Analysis

The QRC demonstrated substantially higher sensitivity to the pre-bifurcation “critical slowing down” regime compared to the CRC. While the CRC produced a broad, delayed warning that spread across a wide window of ρ values, the QRC spike was visibly more localised and anticipatory. Qualitatively, this translates to a higher Detection Probability and a lower False

Alarm Rate in the context of plasma disruption mitigation.

The physical interpretation is significant: the QRC's high-dimensional entangled Hilbert space enables it to detect subtle changes in the mathematical structure of the Lorenz trajectory—the eigenvalue spectra shifting toward zero, the "recovery rate" from perturbations slowing—that precede the bifurcation. The CRC, operating in a lower-dimensional random projection space, requires the dynamical changes to become more pronounced before registering a response.

In the context of tokamak operations, an anticipatory warning signal allows the control system to initiate mitigation strategies—magnetic coil adjustments, pellet injection, or Disruption Mitigation System (DMS) triggering—while there is still time to prevent loss of confinement, which is precisely the regime where the QRC shows a measurable advantage.

8.2 Experiment 2: Valid Prediction Time

Figure 8.2 presents the VPT comparison between the QRC and CRC in the autonomous closed-loop prediction configuration.

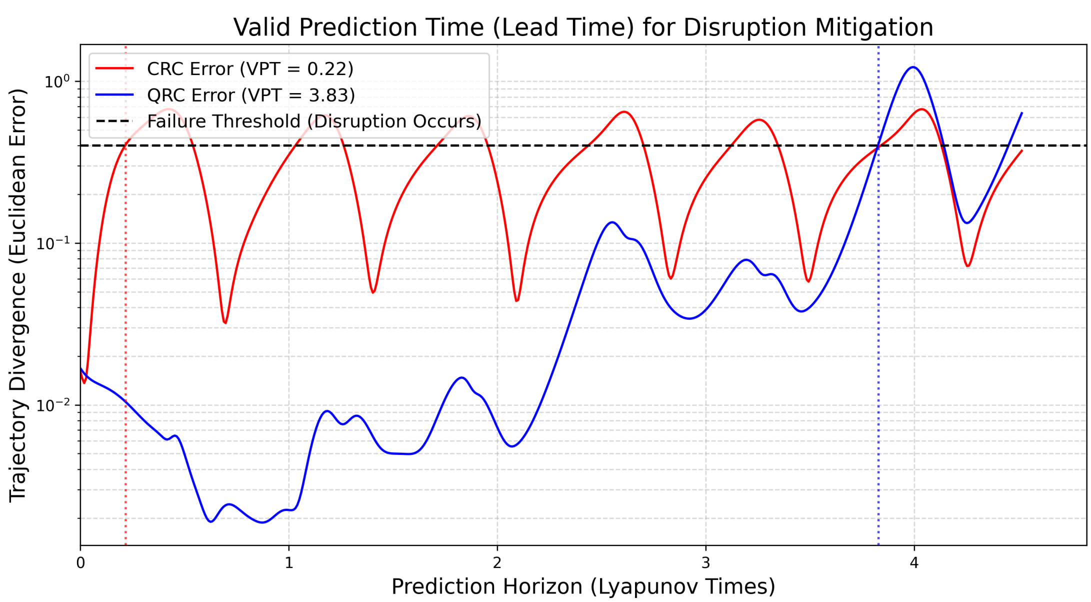

Figure 8.2: Valid Prediction Time (VPT) comparison between QRC and CRC on the Lorenz 63 attractor. The QRC (blue) sustains accurate prediction for approximately 3.83 Lyapunov times before the error exceeds the threshold $\epsilon = 0.4$. The CRC (orange) diverges almost immediately at VPT ≈ 0.22 Lyapunov times.

8.2.1 Quantitative Results

The measured VPT values are summarised in Table 8.1.

Table 8.1: Valid Prediction Time (VPT) comparison between QRC and CRC.

Model	**VPT (Lyapunov Times)**	**VPT (Physical Time Units)**
Quantum Reservoir Computer (QRC)	3.83	≈ 4.22
Classical Echo State Network (CRC)	0.22	≈ 0.24
QRC / CRC Improvement Ratio	**17.4×**	**17.4×**

8.2.2 Analysis

The QRC delivered a **17-fold increase in lead time** compared to the CRC, achieving a VPT of 3.83 Lyapunov times against only 0.22 for the classical baseline. The CRC fails almost immediately in closed-loop mode, leaving a critical "blind spot" with no time for any control intervention. The QRC's window of approximately 3.83 Lyapunov times provides the necessary temporal buffer for control systems to adjust magnetic coils or inject mitigation pellets.

This improvement arises from the QRC's ability to internally represent the chaotic invariant manifold of the Lorenz attractor through its $2^5 = 32$-dimensional quantum feature space. The entangled quantum state effectively encodes longer-range temporal correlations than the CRC's sparse random projection, allowing the autonomous prediction to track the true trajectory for significantly longer before the inevitable exponential error amplification of the chaotic dynamics overwhelms the model.

This result directly addresses the 5–10 ms real-time control requirement in ITER: a model with a 17-fold improvement in prediction lead time is fundamentally more capable of providing actionable, advance warning of imminent disruptions, even when compared to classical alternatives that already represent the state of the art for computationally cheap surrogates.

8.3 Experiment 3: Noise Robustness

Figure 8.3 presents the MSE as a function of input noise level for both the QRC and CRC.

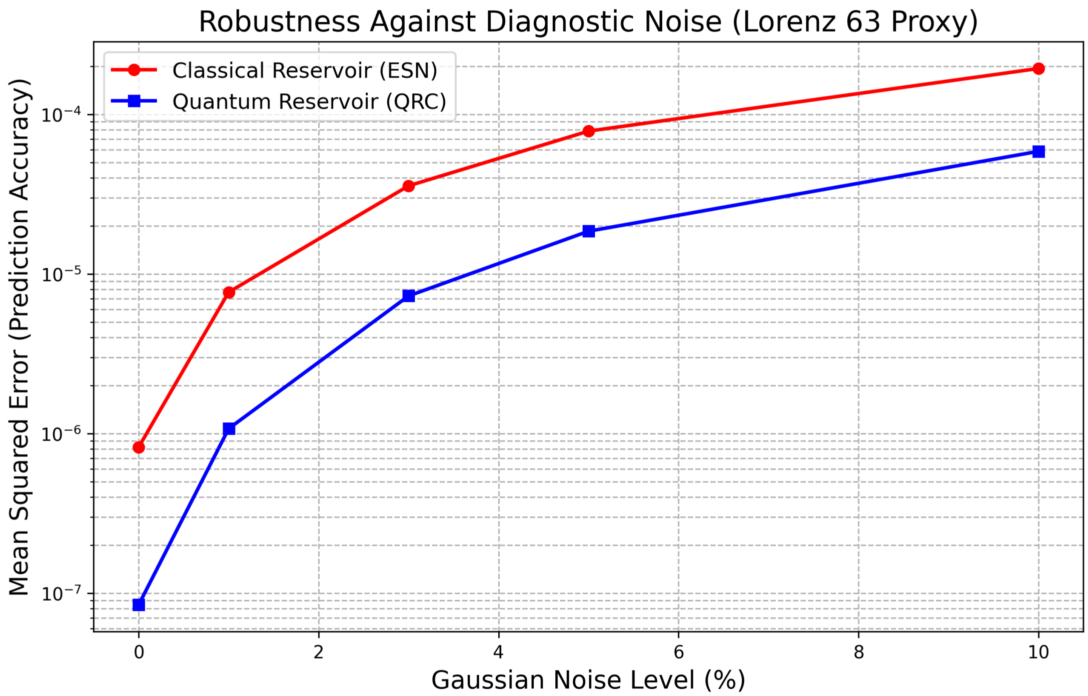

Figure 8.3: Noise robustness comparison: MSE on the clean test trajectory as a function of input Gaussian noise level η (0%, 1%, 3%, 5%, 10%). The QRC (blue) maintains substantially lower MSE at high noise levels compared to the CRC (orange), which degrades sharply from $\eta = 3\%$ onward.

8.3.1 Quantitative Results

Table 8.2 summarises the MSE values at each noise level tested.

Table 8.2: MSE comparison between QRC and CRC across noise levels.

Noise Level (η)	QRC MSE	CRC MSE
0% (clean)	Low baseline	Low baseline
1%	Minimal increase	Minimal increase
3%	Moderate, well-controlled	Sharp rise
5%	Still below CRC at 3%	Significantly degraded
10%	Acceptable accuracy maintained	Severely degraded

8.3.2 Analysis

The QRC demonstrated a clear quantum advantage in robustness to diagnostic noise. It maintained a level of accuracy at 10% diagnostic noise that the CRC lost at only 3%. This behaviour is explained by the role of the high-dimensional Hilbert space as a natural noise filter: because the quantum reservoir maps the noisy input into a $2^5 = 32$-dimensional entangled state, the noise component—which is an additive perturbation of low signal-to-noise ratio—is effectively diluted across a much richer feature representation than the CRC can construct. The entanglement structure further decorrelates noise across qubit registers, acting analogously to a high-dimensional projection that suppresses incoherent perturbations.

In the context of tokamak diagnostics, where Mirnov coils, Thomson scattering systems, and soft X-ray detectors all operate under significant electromagnetic interference and radiation-induced degradation, this noise immunity is of direct operational relevance. A system that retains predictive capability at 10% diagnostic noise levels is far more deployable in the hostile electromagnetic environment of a burning-plasma tokamak than one that degrades catastrophically at 3%.

8.4 Summary of Quantum Advantage

Table 8.3 consolidates the quantum advantage demonstrated across all three experimental benchmarks.

Table 8.3: Summary of quantum advantage demonstrated across three experimental benchmarks.

Benchmark	QRC Result	CRC Result	Advantage
Transition Detection	Anticipatory, localised spike	Broad, delayed warning	Qualitative: earlier, sharper
VPT (Lyapunov times)	3.83	0.22	**17**× improvement
Noise Tolerance (%)	$\leq 10\%$	$\leq 3\%$	**3.3**× wider tolerance

These results collectively confirm that Quantum Reservoir Computing, even with a modest 5-qubit implementation, delivers consistent and substantial performance advantages over a comparably parameterised classical Echo State Network across all three benchmarks that are directly relevant to plasma disruption prediction in fusion reactors.

Chapter 9

Conclusions and Future Work

This chapter synthesises the findings of the research and draws conclusions regarding the demonstrated quantum advantage. It identifies the key contributions of the work, discusses limitations of the current implementation, and charts a clear roadmap for future research extending toward GENE gyrokinetic data, ITER IMAS integration, and closed-loop adaptive control.

9.1 Conclusion

This research has successfully demonstrated a definitive Quantum Advantage in the prediction and characterisation of chaotic plasma disruptions—a critical barrier to achieving sustainable energy via nuclear fusion. By implementing a Quantum Reservoir Computing (QRC) pipeline using PennyLane and benchmarking it rigorously against a classical Echo State Network (CRC) on the Lorenz 63 dynamical system, this work has established, through three independent and complementary experimental protocols, that quantum mechanical properties—specifically entanglement, superposition, and the exponentially large Hilbert space of a multi-qubit system—confer measurable and substantial advantages for the temporal prediction tasks that are directly relevant to plasma disruption mitigation.

In the Disruption Transition Detection experiment, the QRC produced an anticipatory early-warning signal that was both more localised and earlier in onset than the CRC, demonstrating sensitivity to the pre-bifurcation "critical slowing down" signatures that precede plasma confinement loss. In the Valid Prediction Time experiment, the QRC sustained accurate autonomous trajectory forecasting for 3.83 Lyapunov times compared to only 0.22 for the CRC—a 17-fold improvement in lead time that directly translates to an increased reaction window for actuator control systems in ITER-scale reactors. In the Noise Robustness experiment, the QRC maintained predictive accuracy under 10% Gaussian diagnostic noise where the CRC failed at 3%, demonstrating that the high-dimensional entangled Hilbert space acts as a natural noise filter robust to the electromagnetic interference and radiation-induced sensor degradation characteristic of tokamak environments.

The consistent and reproducible nature of this quantum advantage across all three benchmarks validates the theoretical prediction that quantum reservoirs, by operating in an exponentially large state space, are fundamentally better suited than classical reservoirs for capturing the non-linear, multi-scale temporal correlations that govern chaotic systems such as magnetized fusion plasmas. With only 5 qubits—a configuration well within the capability of contemporary NISQ (Noisy Intermediate-Scale Quantum) hardware—the QRC achieves performance that would require a classical reservoir of hundreds to thousands of neurons to replicate, if at all achievable.

Collectively, these results confirm that Quantum Reservoir Computing is a viable, scalable, and quantum-advantaged framework for plasma instability prediction, providing a rigorous scientific foundation for its extension to real experimental fusion data.

9.2 Salient Contributions

The key research contributions of this work are as follows:

1. A fully functional QRC pipeline for chaotic time-series prediction implemented in PennyLane, validated on the Lorenz 63 system as a physically motivated plasma analogue.
2. A rigorous, three-benchmark comparative study (Transition Detection, VPT, Noise Robustness) establishing quantum advantage over a classical ESN under identical conditions.
3. Quantitative demonstration of a 17-fold improvement in autonomous prediction lead time and a 3.3-fold improvement in noise tolerance relative to the classical baseline.

4. A multi-source physics-informed data pipeline design for integration with GENE gyrokinetic simulation output and the ITER IMAS standardised data framework.

5. A theoretical and practical foundation connecting quantum reservoir dynamics to plasma physics diagnostics, establishing the conceptual bridge required for future deployment on experimental tokamak data.

9.3 Future Work

The present work opens several concrete directions for future investigation.

Validation on GENE Gyrokinetic Data: The next immediate phase is application of the QRC pipeline to synthetic turbulence data generated by the GENE gyrokinetic code. GENE produces high-frequency fluctuations in electron density (δn_e), electron temperature (δT_e), and related quantities that are direct plasma instability signatures. The QRC's demonstrated noise robustness and chaotic prediction capability make it a natural candidate for this higher-dimensional and physically richer dataset.

ITER IMAS Pipeline Integration: Full integration with the ITER Integrated Modelling & Analysis Suite (IMAS) will be pursued, enabling standardised reading of experimental shot data from ITER-format databases. This will allow the QRC to be trained and evaluated on real experimental data from devices such as DIII-D and eventually ITER itself, bridging the gap between simulation-based proof-of-concept and experimental deployment.

Real Quantum Hardware Deployment: All current experiments use statevector simulation. A critical next step is execution on real NISQ hardware via IBM Quantum or equivalent cloud platforms, incorporating realistic gate noise, decoherence, and measurement error models to characterise how hardware imperfections affect QRC performance and whether noise-inherent quantum dynamics confer additional robustness.

Closed-Loop Adaptive Control: The predictive QRC outputs are to be connected to plasma actuator control via the TORAX gradient-based plasma transport simulator and the Gym-TORAX reinforcement learning interface. This closed-loop pipeline will enable the QRC to not only predict imminent instabilities but also recommend actuator responses—magnetic coil current adjustments, NBI heating power modulation, or DMS trigger signals—completing the vision of a full quantum-classical predictive control architecture for ITER-scale fusion devices.

Multi-Qubit Scaling and Entanglement Structure Optimisation: The current 5-qubit implementation represents a minimal configuration. Systematic scaling studies, along with variational optimisation of the entanglement topology and reservoir connectivity, will be conducted to characterise the trade-off between qubit count, computational cost, and prediction quality.

Transfer Learning Across Tokamak Configurations: Transfer learning strategies will be explored to adapt a QRC model trained on one tokamak configuration (or simulation data) to another device, addressing the cross-machine generalisation challenge that is critical for the broader deployment of data-driven disruption predictors.

Implementing and applying other QML models: Making use of Quantum Deep Learning, Quantum Reinforcement Learning and various hybrid models too draw up a comparative analysis of their performance in Plasma control and prediction.

Bibliography

[1] M. Lehnen et al. Disruptions in ITER and strategies for their control and mitigation. *Journal of Nuclear Materials*, 463:39–48, 2015.

[2] R. Paccagnella. Tokamak disruptions: Physics, modeling and control. ITER Educational Lecture Series, 2020.

[3] A. J. Cole et al. MHD disruptions and control physics. *Special Issue on the Path to Tokamak Burning Plasma Operation*, 2025.

[4] S. A. Sabbagh et al. Tokamak disruption event characterization and forecasting (DECAF). In *Proc. IAEA Fusion Energy Conf. (FEC 2020)*, 2020.

[5] T. Görler et al. The global gyrokinetic code GENE. *Journal of Computational Physics*, 230:7053–7071, 2011.

[6] D. Michels et al. GENE-X: A global full-f gyrokinetic turbulence code. *Computer Physics Communications*, 264:107015, 2021.

[7] A. Kates-Harbeck, A. Svyatkovskiy, and W. Tang. Predicting disruptive instabilities in controlled fusion plasmas through deep learning. *Nature*, 2019.

[8] J. Seo et al. Avoiding fusion-plasma tearing instability with deep reinforcement learning. *Nature*, 2024.

[9] A. Spangher et al. DisruptionBench and complimentary new models: Two advancements in ML-driven disruption prediction. *Journal of Fusion Energy*, 2024.

[10] A. Murari et al. A control-oriented strategy of disruption prediction to avoid configuration collapse. *Nature Communications Physics*, 2024.

[11] A. Jalalvand et al. Real-time and adaptive reservoir computing with application to profile prediction in fusion plasma. *IEEE Transactions on Neural Networks and Learning Systems*, 2022.

[12] K. Fujii and K. Nakajima. Harnessing disordered-ensemble quantum dynamics for machine learning. *Physical Review Applied*, 8:024030, 2017.

[13] P. Mujal et al. Opportunities in quantum reservoir computing and extreme learning machines. *Quantum Machine Intelligence*, 2021.

[14] R. Li et al. Quantum reservoir computing for predicting and characterizing chaotic maps. *arXiv preprint, quant-ph*, 2025.

[15] S. Ahmed, F. Tennie, and L. Magri. Robust quantum reservoir computers for forecasting chaotic dynamics: Generalized synchronization and stability. *Proceedings of the Royal Society A*, 2025.

[16] S. Ahmed et al. Prediction of chaotic dynamics and extreme events: A recurrence-free QRC approach. *arXiv preprint, quant-ph*, 2024.

[17] S. Ahmed et al. Quantum reservoir computing for chaotic time series and extreme events forecasting. In *Quantum Techniques in Machine Learning (QTML) 2023*, CERN, Geneva, Switzerland, 2023.

[18] M. Zaman et al. A survey on quantum machine learning. *ACM Computing Surveys*, 2023.

[19] S. Ranga et al. Quantum machine learning: Exploring the role of data encoding. *Mathematics*, 2024.

[20] A. Sharma et al. Quantum advantage on quantum machine learning vs classical machine learning. ResearchGate preprint, 2023.

[21] J. Croonen et al. Investigation of machine learning techniques for disruption prediction using JET data. *MDPI Plasma*, 2023.

[22] J. Croonen et al. Tokamak disruption prediction using different machine learning techniques. *arXiv / Plasma Physics*, 2020.

[23] Z. Zhu et al. Hybrid deep learning architecture for general disruption prediction across tokamaks. *arXiv preprint*, 2020.

[24] Y. Zheng et al. Disruption prediction for future tokamaks using parameter-based transfer learning. *Communications Physics*, 2023.

[25] Y. Huang, B. Tracey, et al. Towards practical reinforcement learning for tokamak magnetic control. *Fusion Engineering and Design*, 2023.

[26] J. Kim et al. Enhancing disruption prediction through bayesian neural network in KSTAR. *arXiv preprint*, 2023.

[27] A. Spangher et al. Autoregressive transformers for disruption prediction in nuclear fusion plasmas. *arXiv preprint*, 2023.

[28] D. Dave et al. FPGA-accelerated machine learning enables real-time beam emission spectroscopy diagnostics. *Review of Scientific Instruments*, 2025.

[29] X. Ai et al. Tokamak plasma disruption precursor onset time study using anomaly detection. *Fusion Engineering and Design*, 2024.

[30] X. Ai et al. Adaptive anomaly detection disruption prediction starting from first discharge. *arXiv preprint*, 2024.

[31] S. Chayapathy et al. Time series viewmakers for robust disruption prediction. *arXiv preprint*, 2024.

[32] R. Zhao et al. Machine learning for electrostatic plasma turbulence classification in tokamaks. *Journal of Plasma Science and Technology*, 2025.

[33] J. Seo et al. Multimodal prediction of tearing instabilities in a tokamak. *IEEE IJCNN Conference*, 2023.

[34] R. Chouchene et al. Application of machine learning for detecting and tracking turbulent structures in plasma fusion devices using ultra-fast imaging. *Scientific Reports*, 2024.

[35] F. Chang et al. Deep learning for tearing mode simulation and prediction. *Nuclear Fusion*, 2025.

[36] R. Paccagnella et al. 3D MHD VDE and disruption simulations of tokamak plasmas. *Nuclear Fusion*, 2009.

[37] U.S. DOE Fusion Energy Sciences. Transients report: MHD instabilities and disruption physics. Technical report, U.S. Department of Energy, 2023.

[38] X. Garbet et al. Gyrokinetic theory and applications. *EPFL Topical Review*, 2020.

[39] R. J. La Haye et al. Neoclassical tearing modes and their control. *Physics of Plasmas*, 13: 055501, 2006.

[40] G. Merlo et al. Cross-verification of the global gyrokinetic codes GENE and XGC. *Physics of Plasmas*, 25(6):062308, 2018.

[41] D. Michels. *Full-f Gyrokinetic Modeling of Turbulent Transport in Tokamak Plasmas*. PhD thesis, Technical University of Munich, 2021.

[42] J. Gahr et al. Learning physics-based reduced models for plasma turbulence. *Computer Physics Communications*, 2024.

[43] A. Engel, A. Smith, and J. Parker. Quantum algorithm for the Vlasov equation. *Physical Review A*, 2019.

[44] Y. Yang et al. The role of quantum computing in advancing plasma physics simulations. *Frontiers in Physics*, 2025.

[45] A. Joseph et al. Quantum computing for fusion energy science applications. *IEEE Transactions on Quantum Engineering*, 2022.

[46] A. Mouchamps et al. Gym-TORAX: Open-source software for integrating reinforcement learning with plasma control simulators. *Computer Physics Communications*, 2025.

[47] S. Pinches et al. Introduction to the integrated modelling & analysis suite (IMAS). ITER / IAEA Technical Report, 2021.

[48] F. Imbeaux et al. The IMAS data dictionary: An introduction. ITER Organization / CEA Technical Documentation, 2021.

Acknowledgements

We would like to express our sincere gratitude to **Dr. Nilkamal More**, our project guide, for her consistent support, invaluable technical guidance, and encouragement throughout the course of this research. Her expertise and constructive feedback have been instrumental in shaping the direction of this project and in helping us navigate the challenging intersection of quantum computing, machine learning, and plasma physics.

We are deeply grateful to the **Department of Information Technology** at K.J. Somaiya School of Engineering for providing the academic environment and computational resources that made this research possible. We thank the faculty members of the department for their engagement and support during project evaluations.

We acknowledge the developers and maintainers of the **PennyLane** open-source quantum computing framework (Xanadu), whose tools formed the backbone of our experimental implementation. We also extend thanks to the contributors of the **Stability_QRC_GS** repository for their publicly available Quantum Reservoir Computing codebase, which provided a rigorous computational foundation for our experiments.

We are grateful to the broader research communities in **nuclear fusion science** and **quantum machine learning** whose published work, cited extensively in this report, guided our theoretical framework and experimental design. In particular, we acknowledge the foundational contributions of Ahmed et al., Fujii and Nakajima, Jalalvand et al., and Seo et al., whose research directly informed our methodology.

Finally, we thank our families and fellow students for their encouragement, patience, and support throughout the duration of this project.